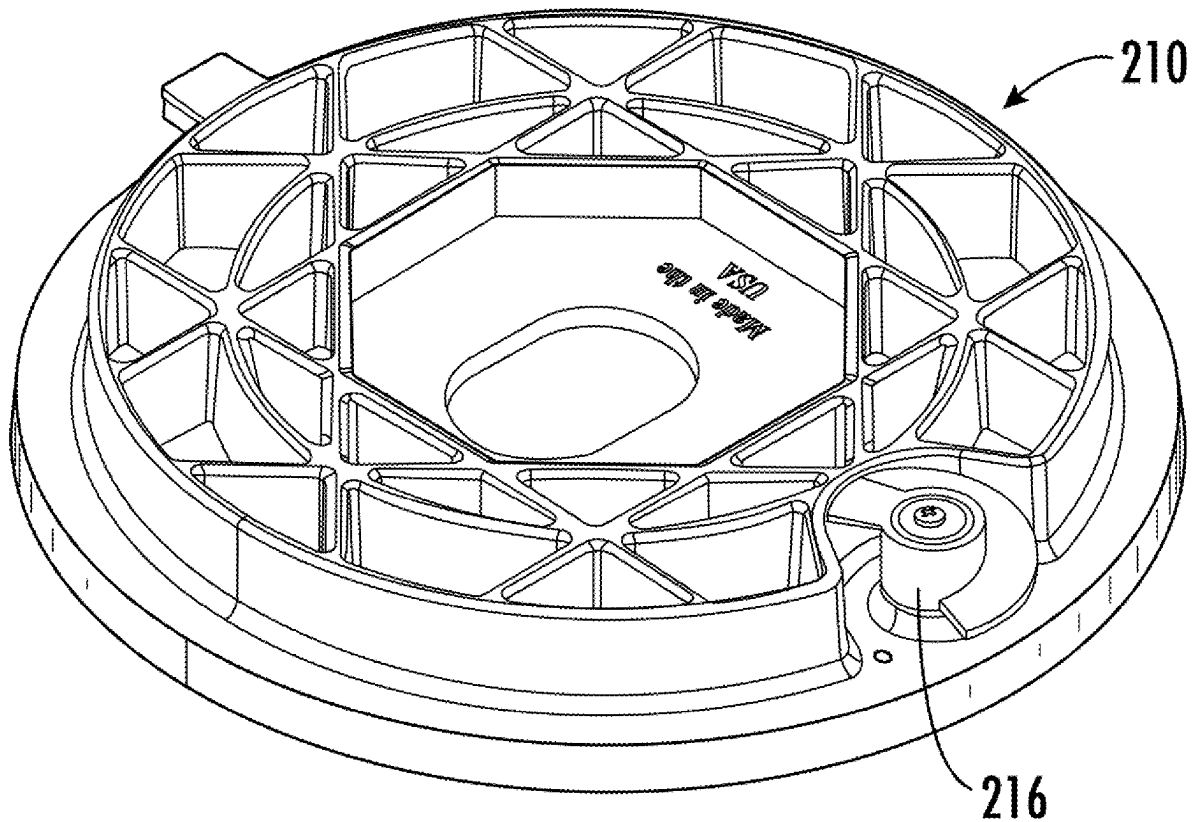




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Butcher et al.(10) **Pub. No.: US 2025/0270006 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **WATER OR UTILITY METER BOX LID
SUITABLE FOR MULTIPLE TYPES OF
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CT (US)(21) Appl. No.: **18/813,425**(22) Filed: **Aug. 23, 2024****Related U.S. Application Data**(60) Provisional application No. 63/556,778, filed on Feb.
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B65D 43/02 (2006.01)
(52) **U.S. Cl.**
CPC **B65D 43/26** (2013.01); **B65D 43/0202**
(2013.01); **B65D 2543/00092** (2013.01)(57) **ABSTRACT**

Systems and methods are disclosed for a lid for covering a utility or meter box. The water or utility meter box lid is particularly suitable for having installed thereon multiple different types of latch mechanisms, e.g., worm latches, spring-loaded latches, and the like. The lid includes a top side, a bottom side, and a through-hole extending from the top side to the bottom side into which a selected latch type may be installed. The bottom surface of the lid has a surface for rotatably engaging a latch portion of a latch mechanism, and may also include at least one opening such as a cavity or retainer, that engages an end of a spring, allowing a user to select and install, into a single type of lid, either one of a worm latch or a spring-loaded latch.



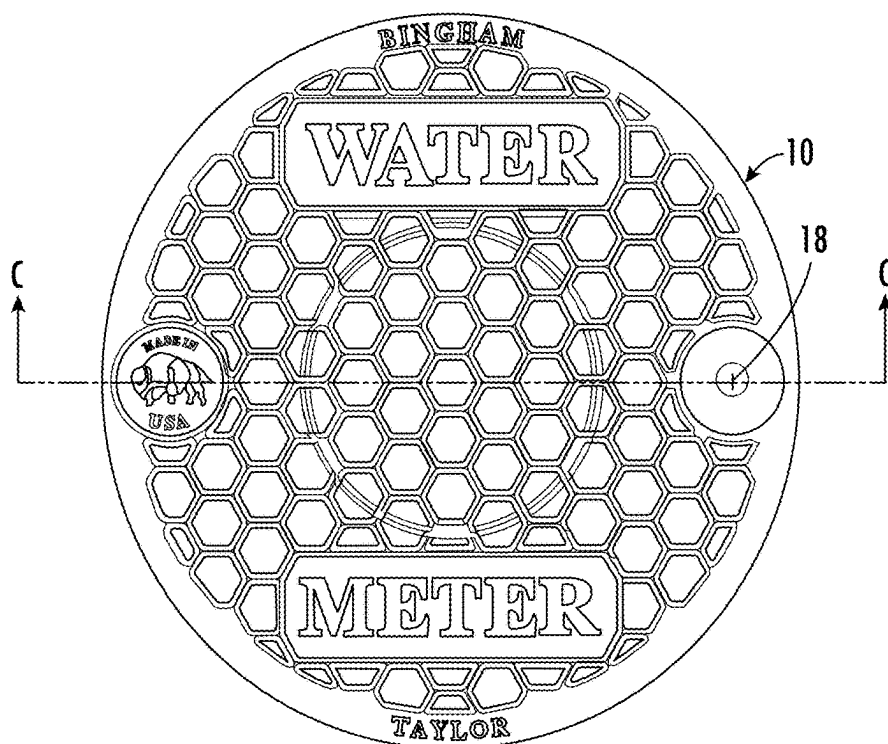


FIG. 1A

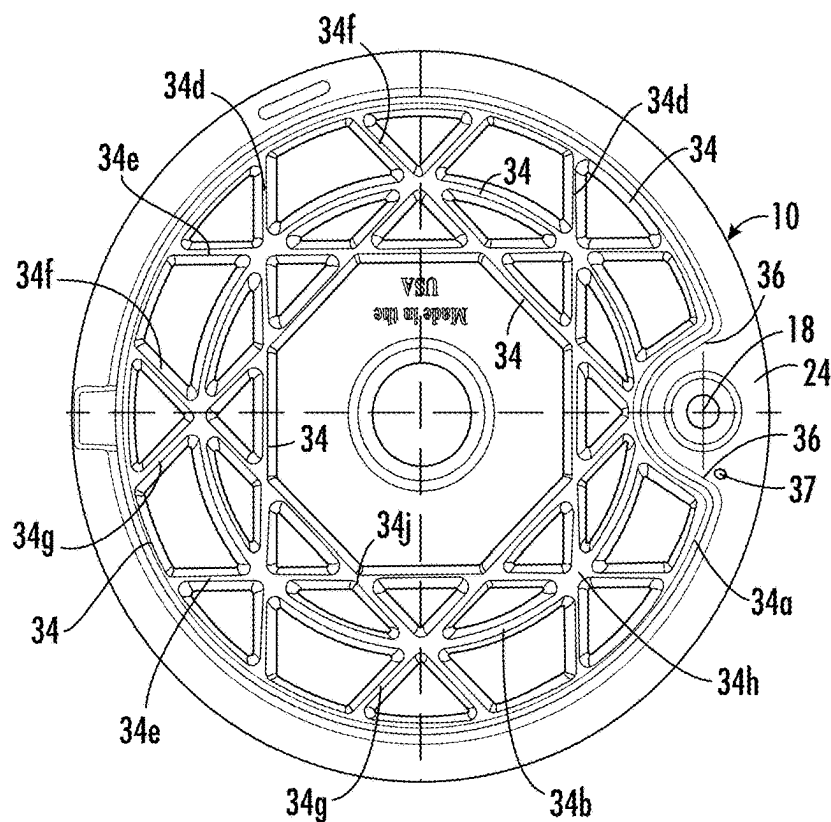


FIG. 1B

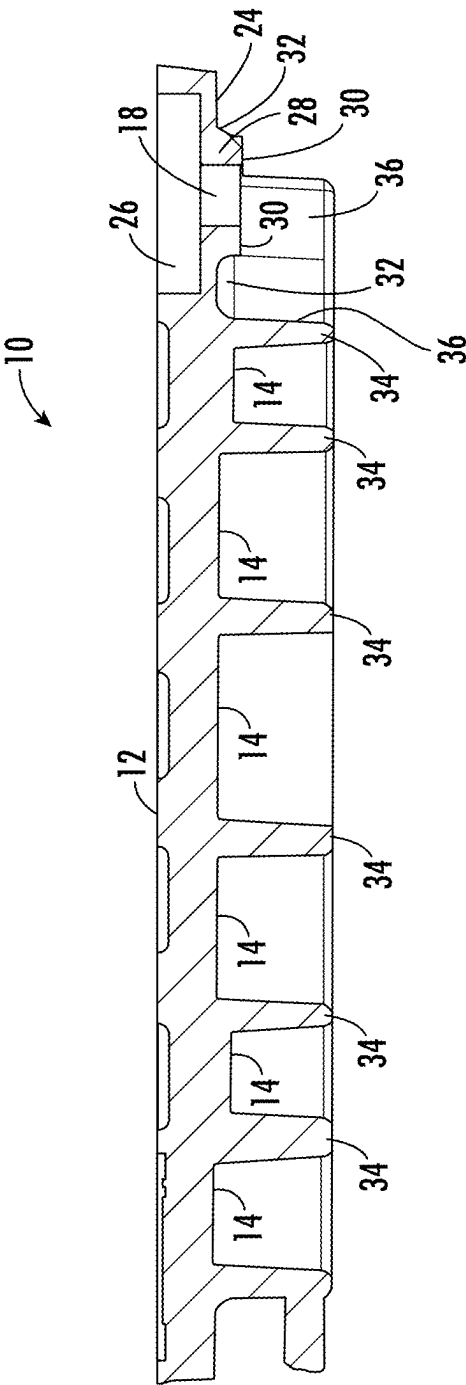


FIG. 1C

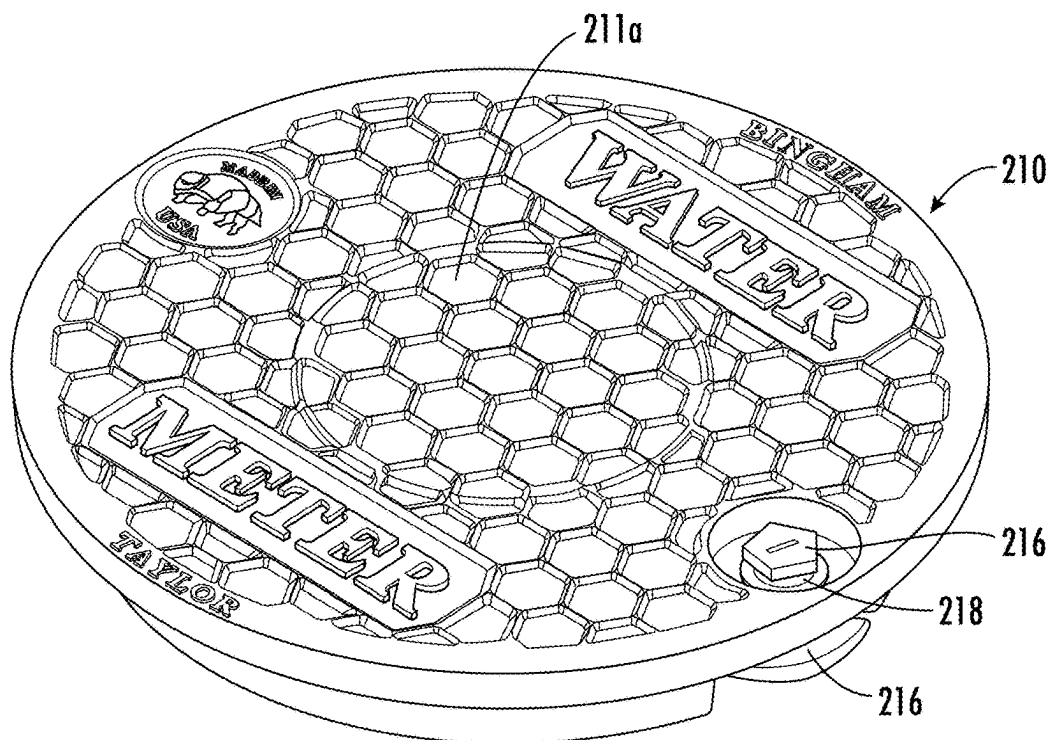


FIG. 2A

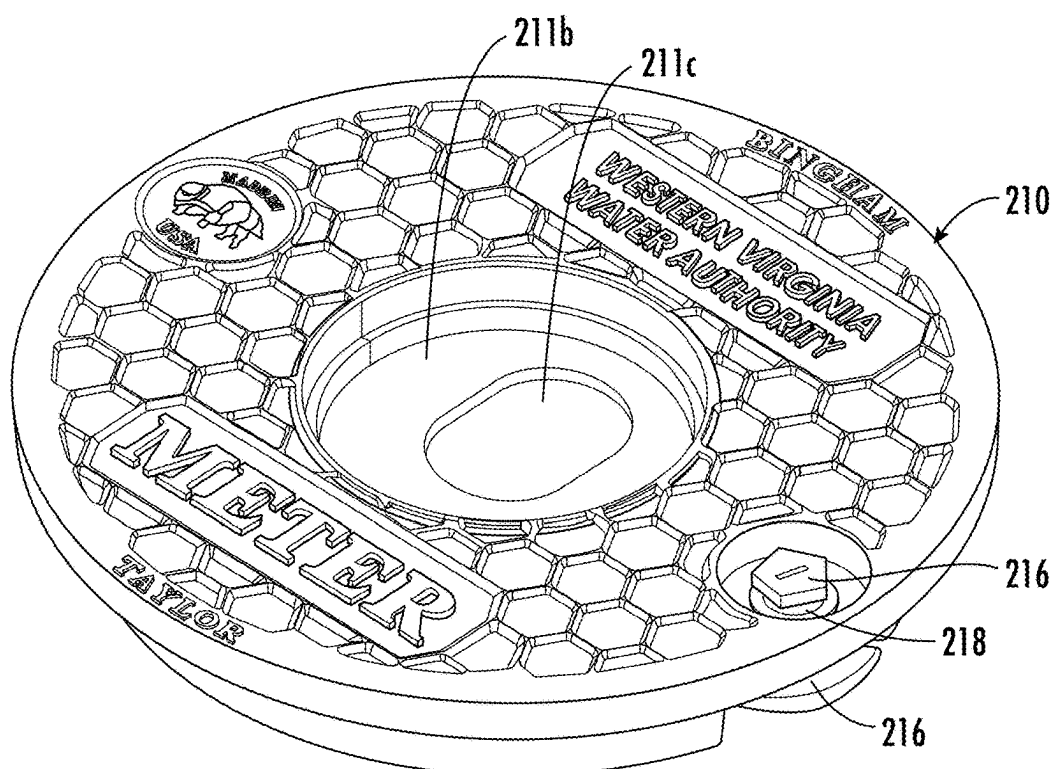


FIG. 2B

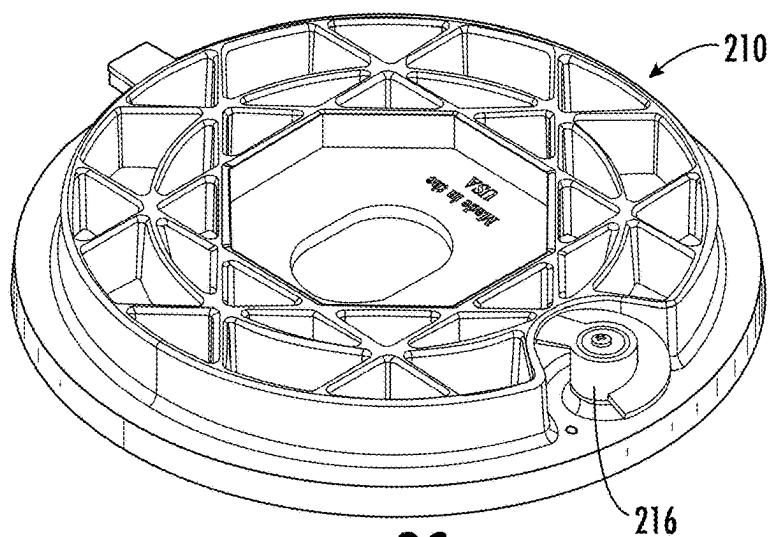


FIG. 2C

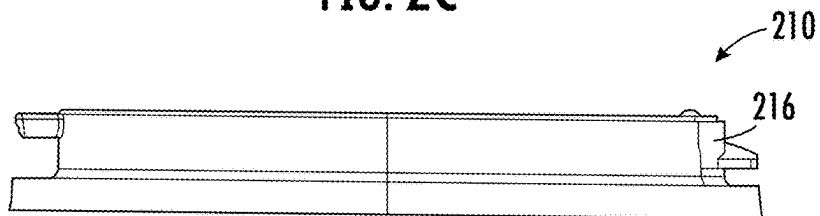


FIG. 2D

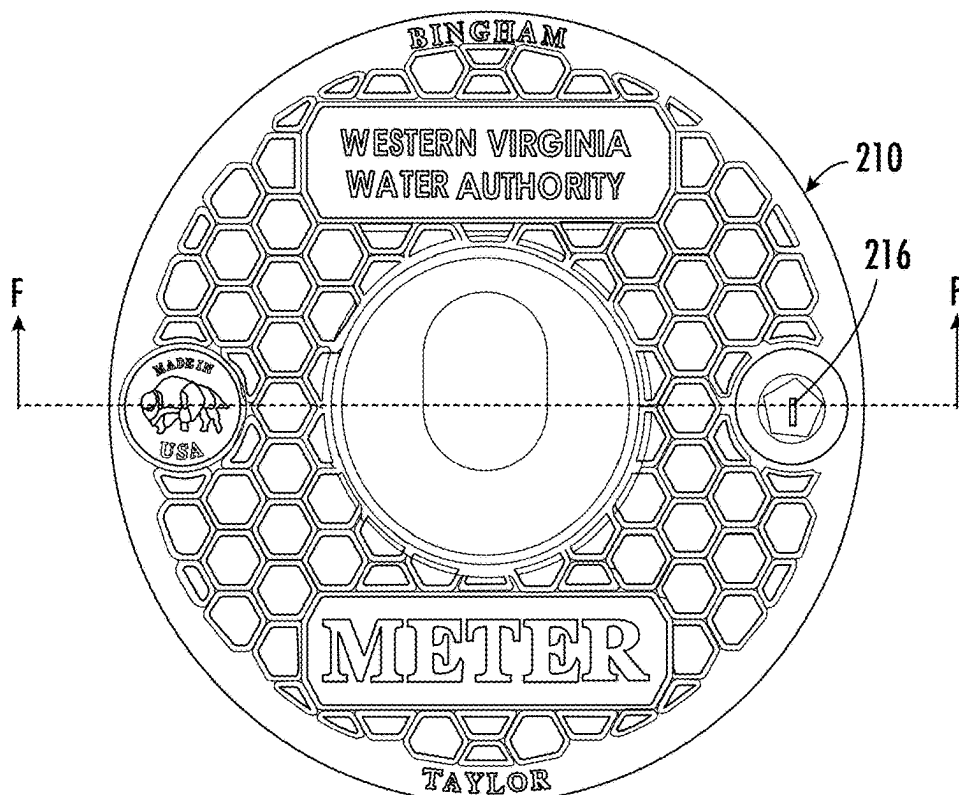


FIG. 2E

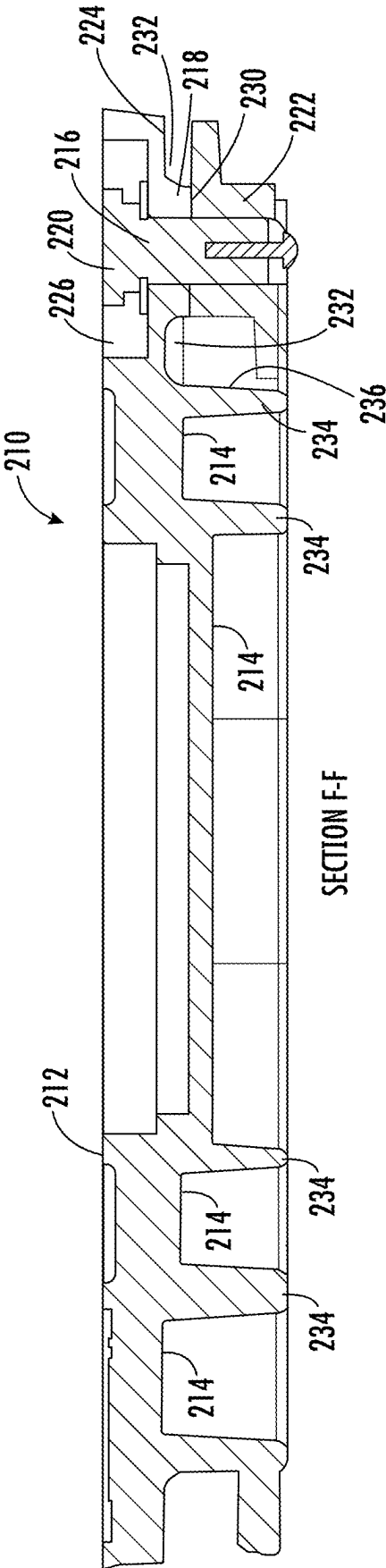


FIG. 2F

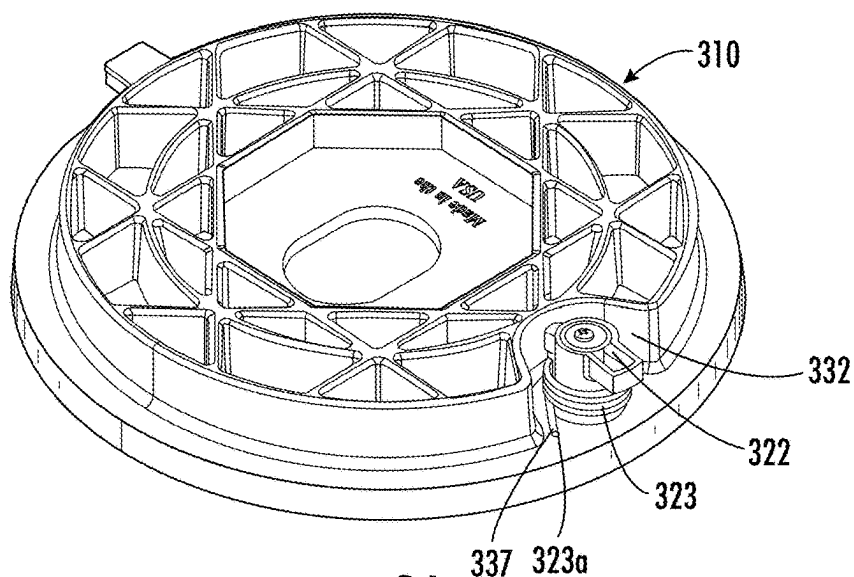


FIG. 3A

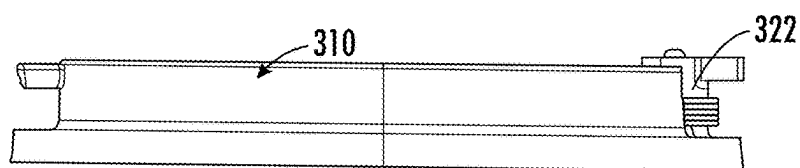


FIG. 3B

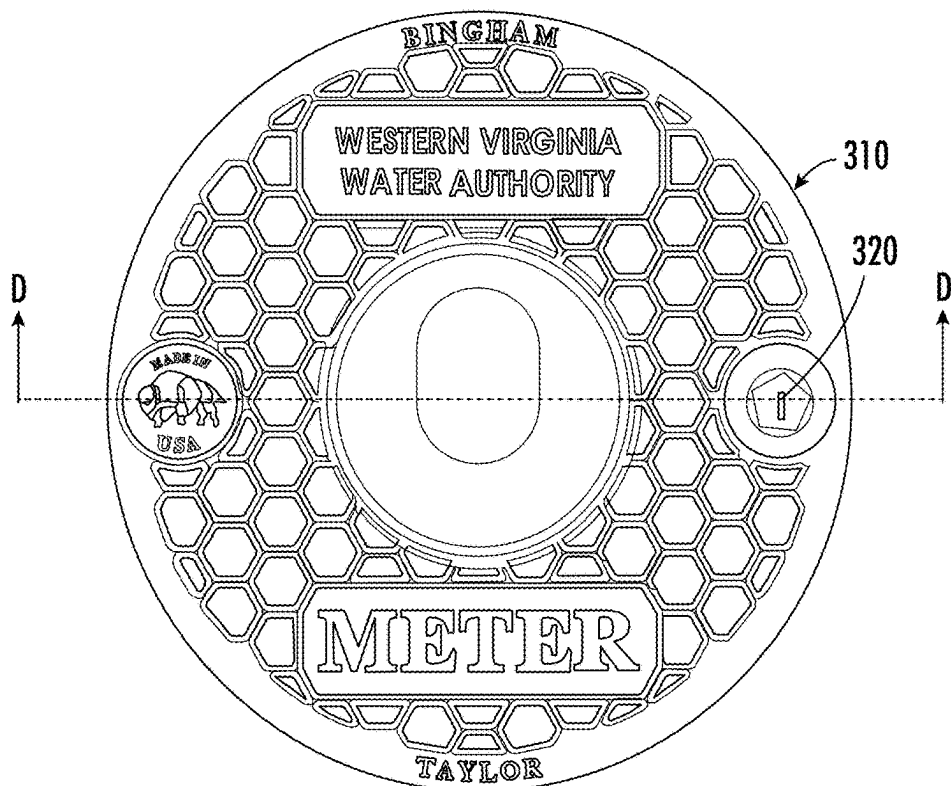


FIG. 3C

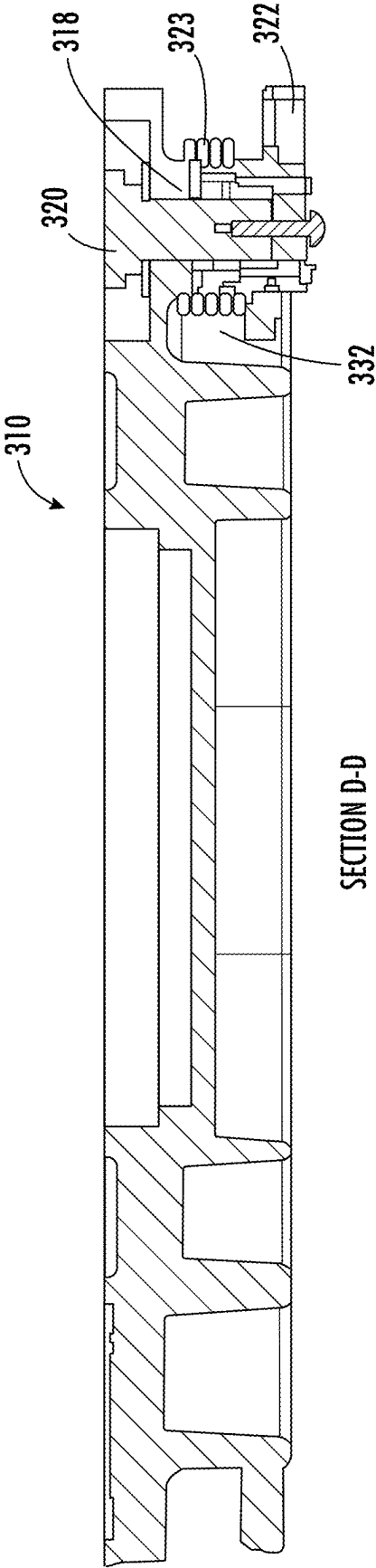


FIG. 3D

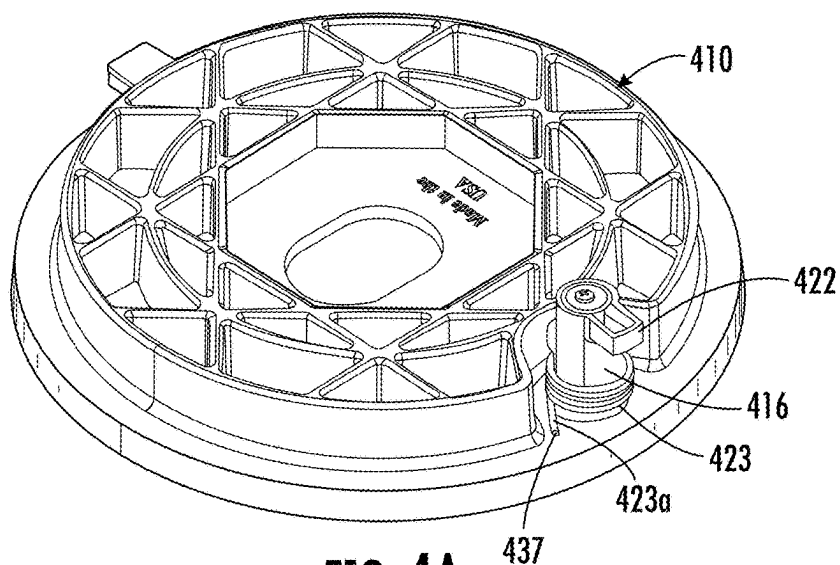


FIG. 4A

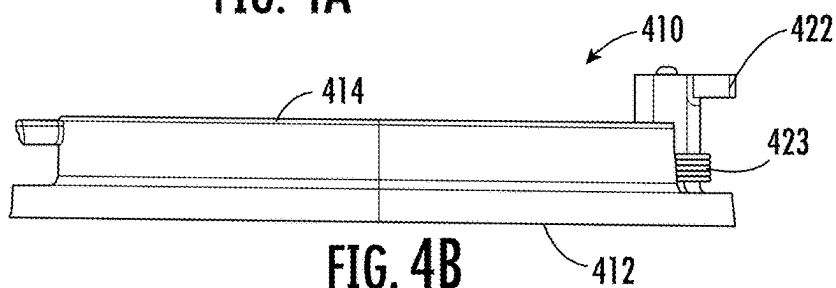


FIG. 4B

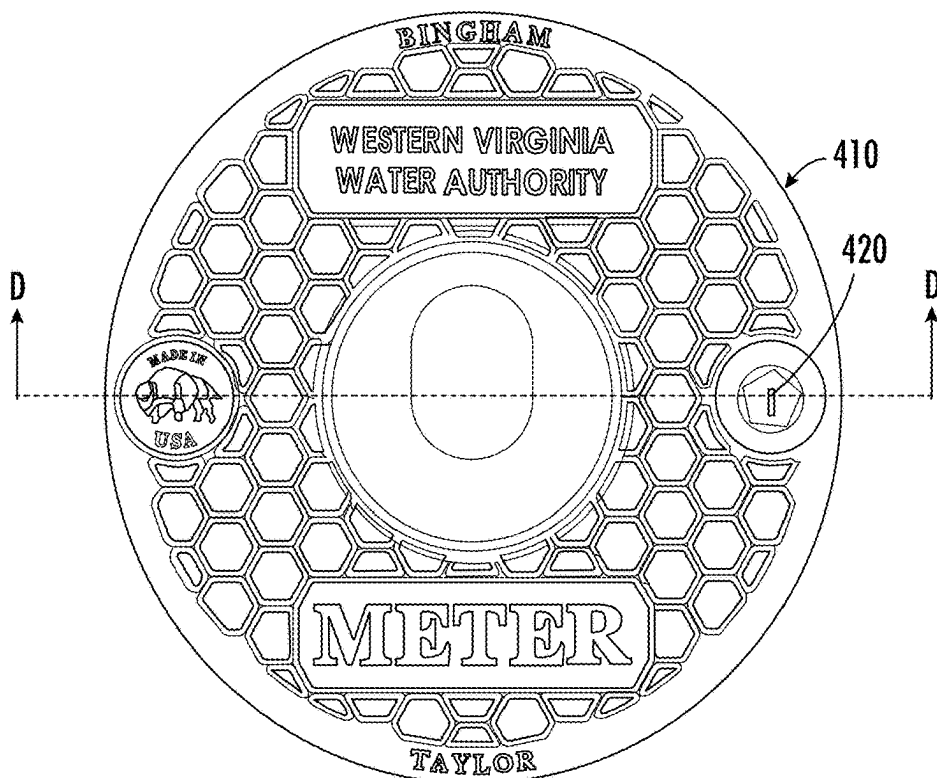


FIG. 4C

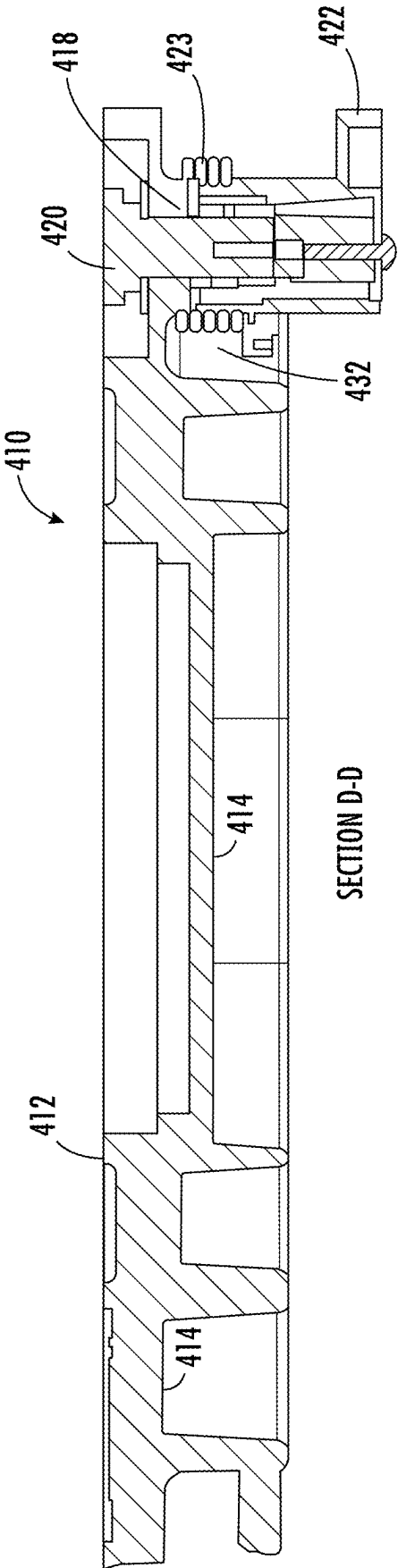


FIG. 4D

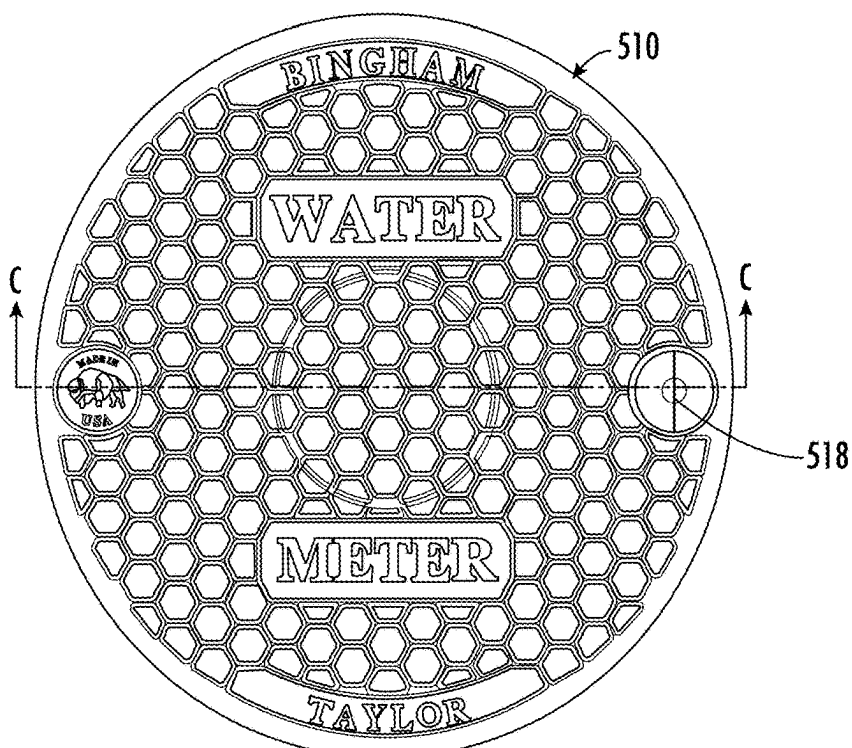


FIG. 5A

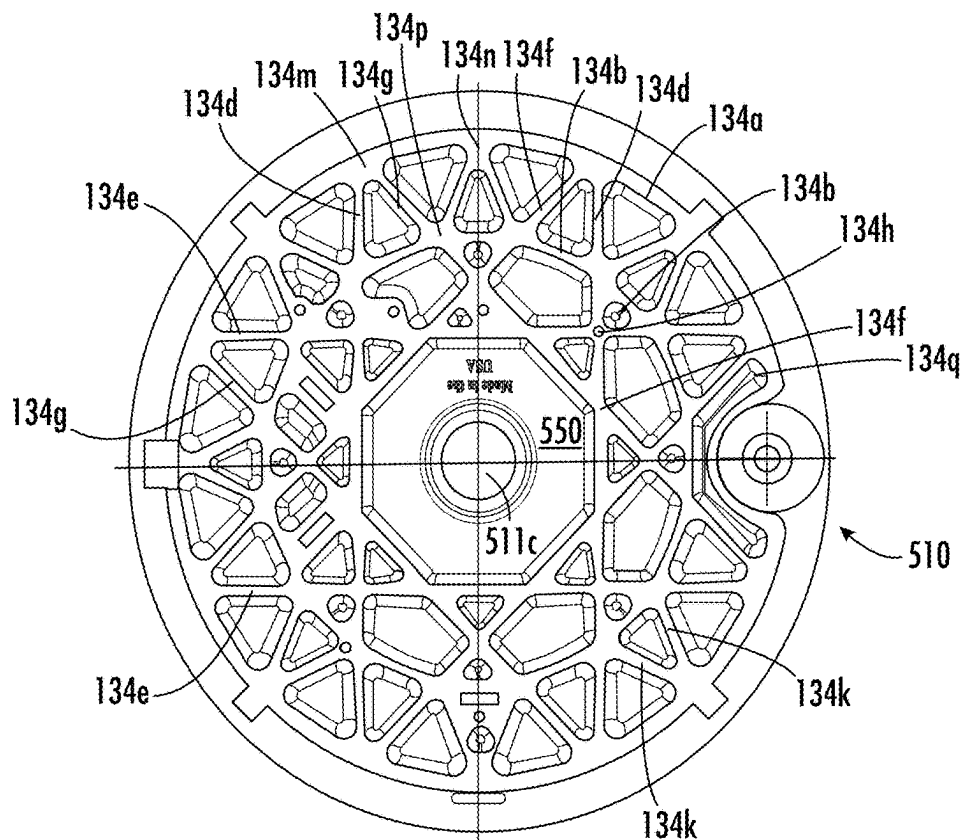
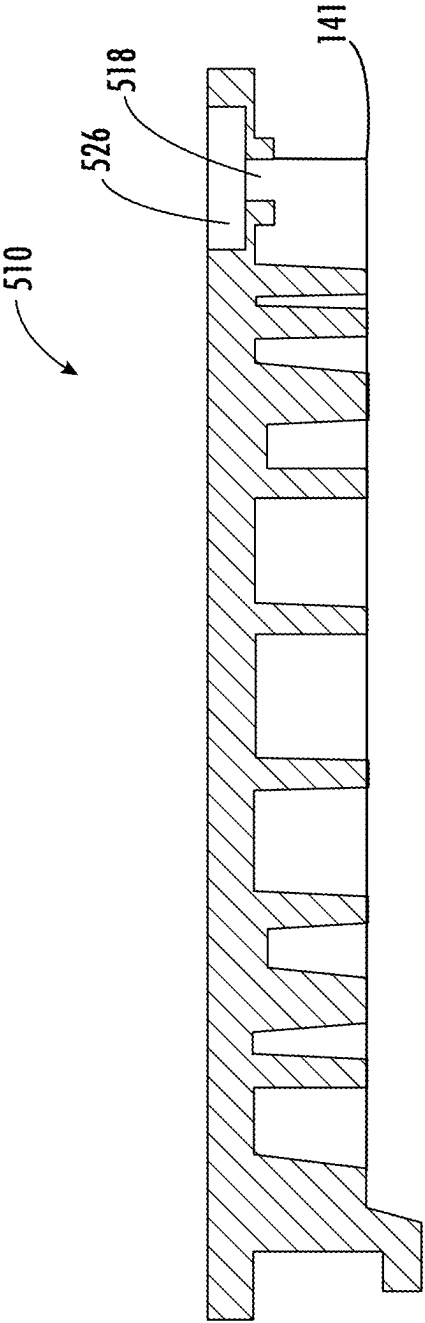


FIG. 5B



SECTION C-C

FIG. 5C

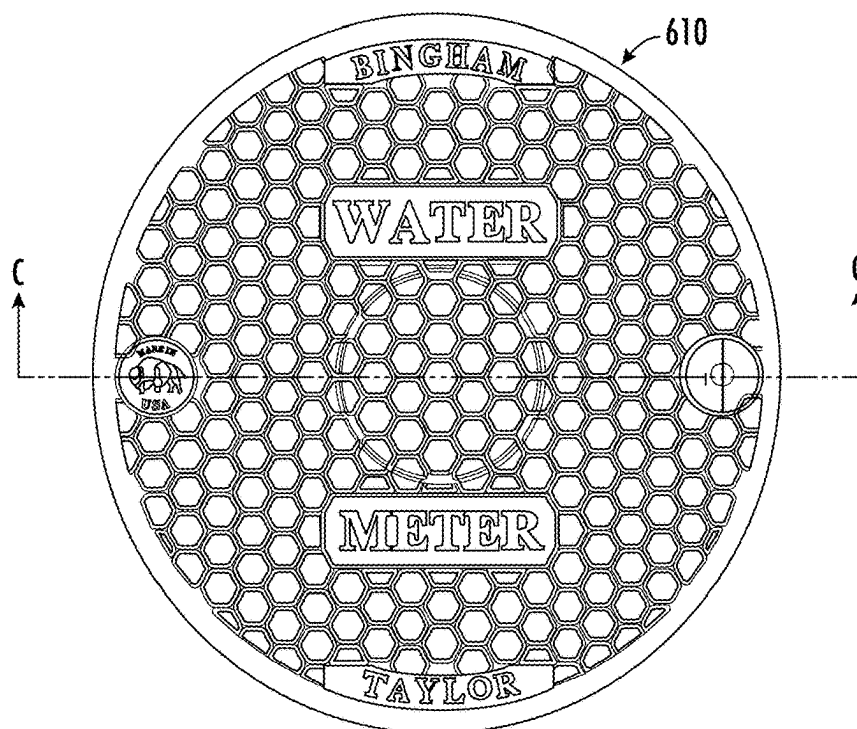


FIG. 6A

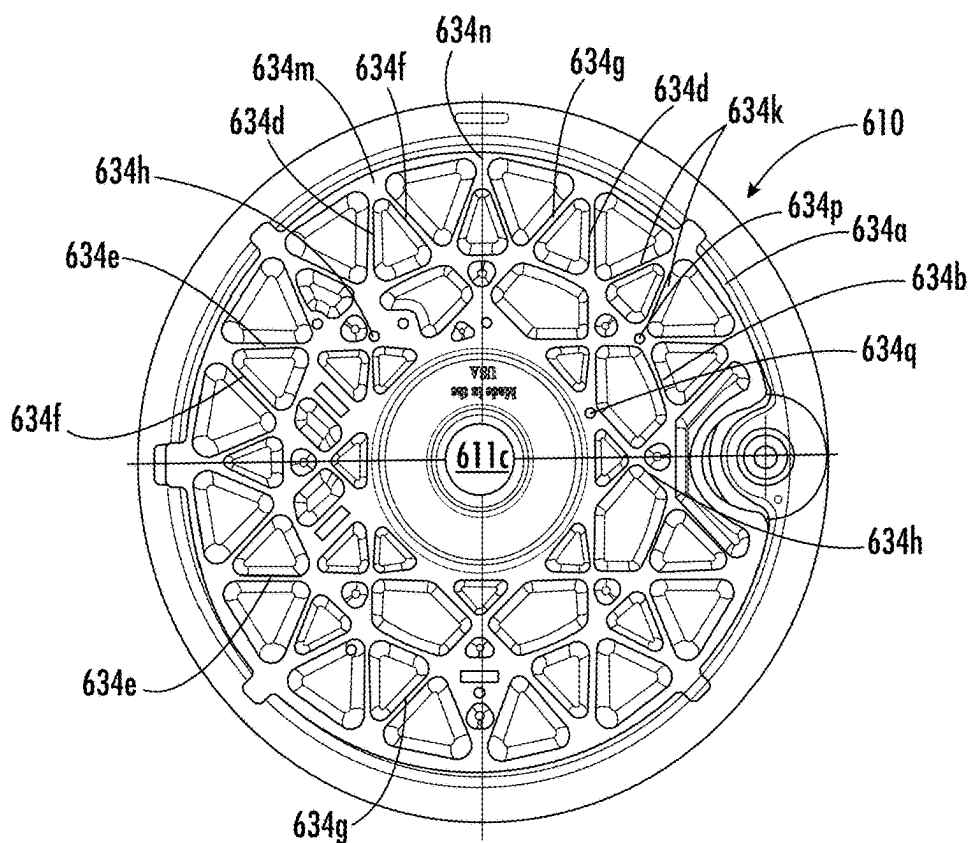
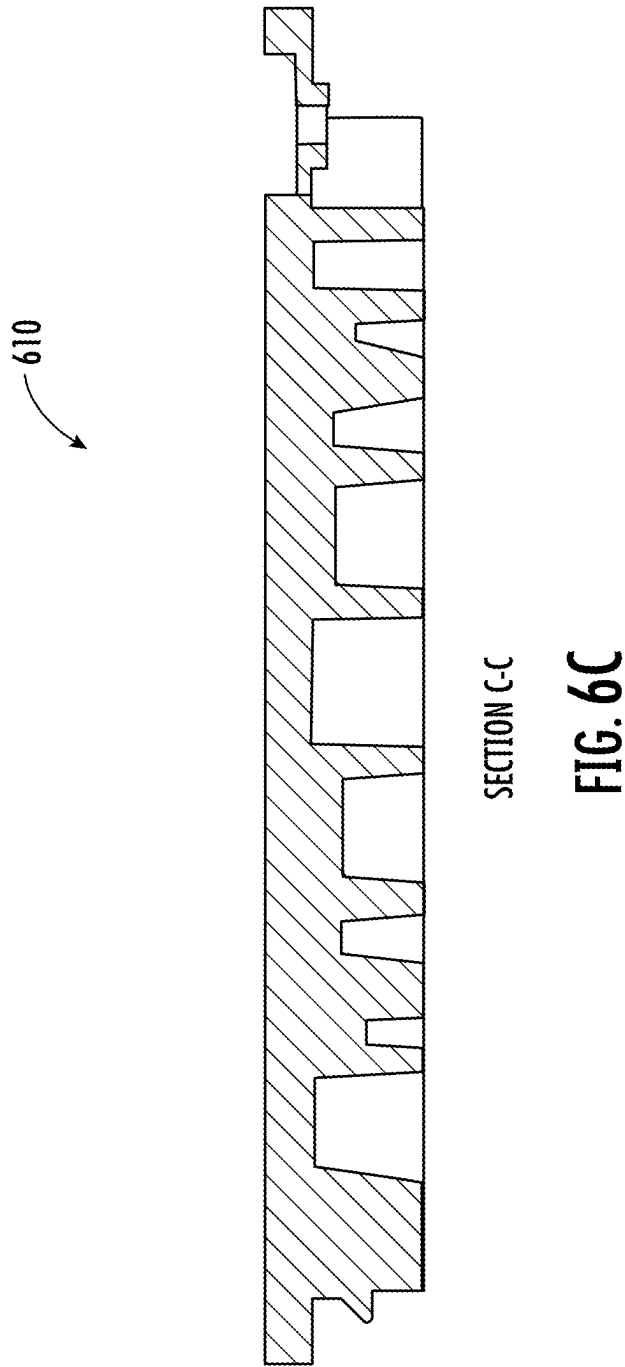


FIG. 6B



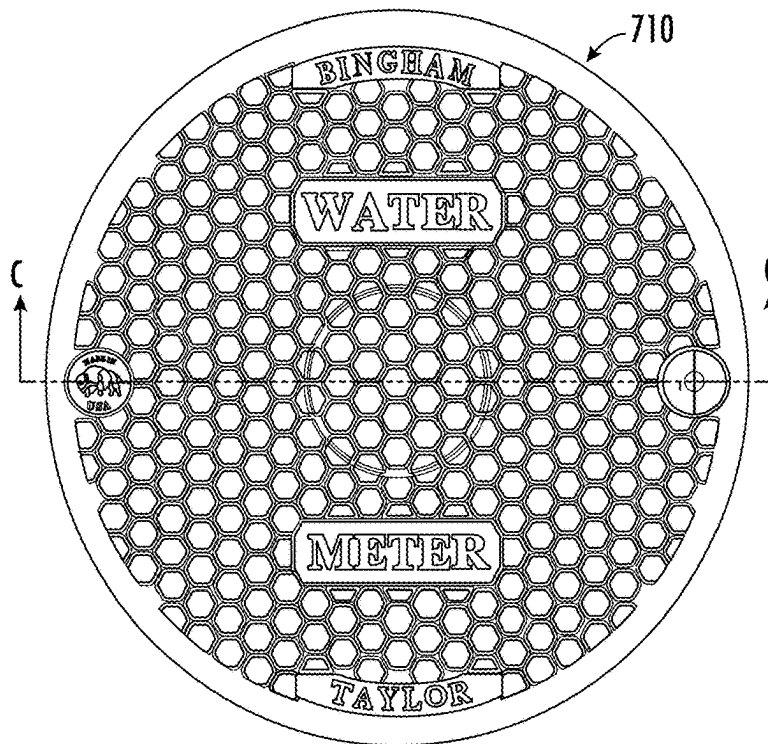


FIG. 7A

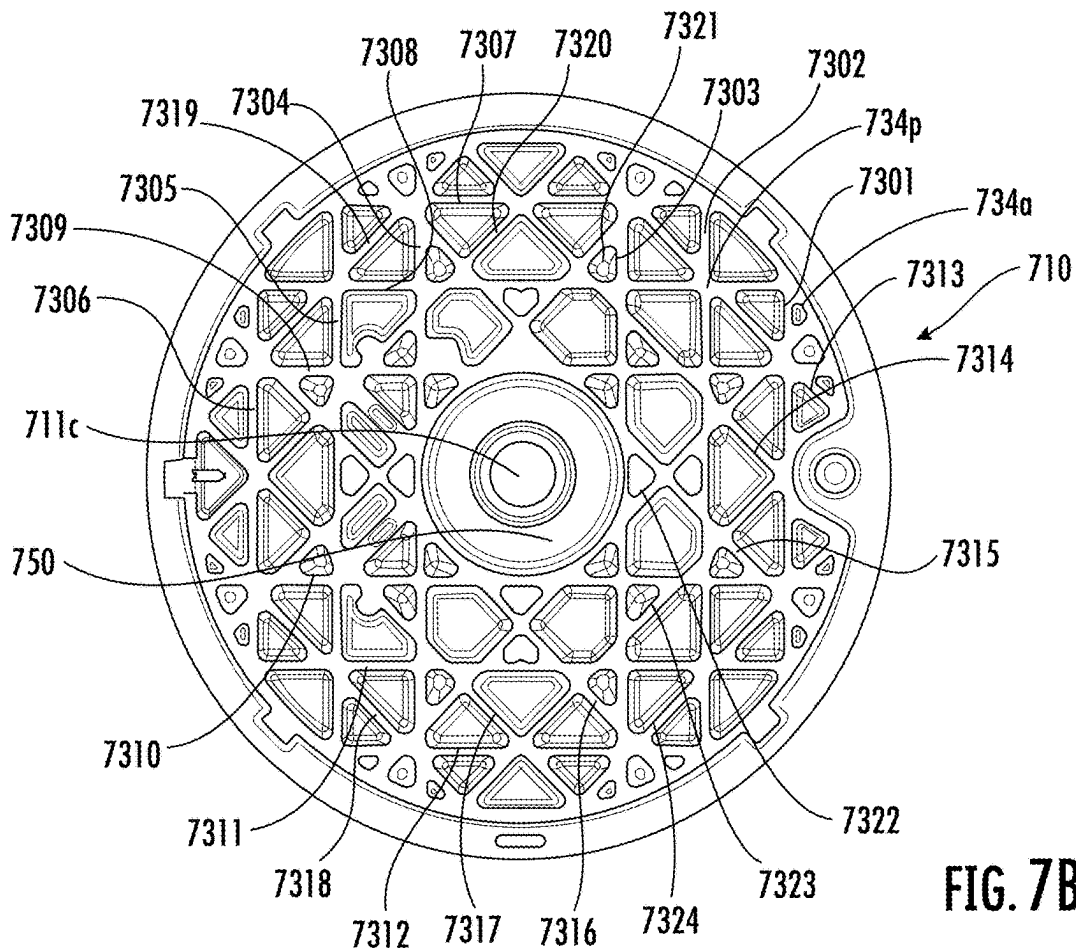
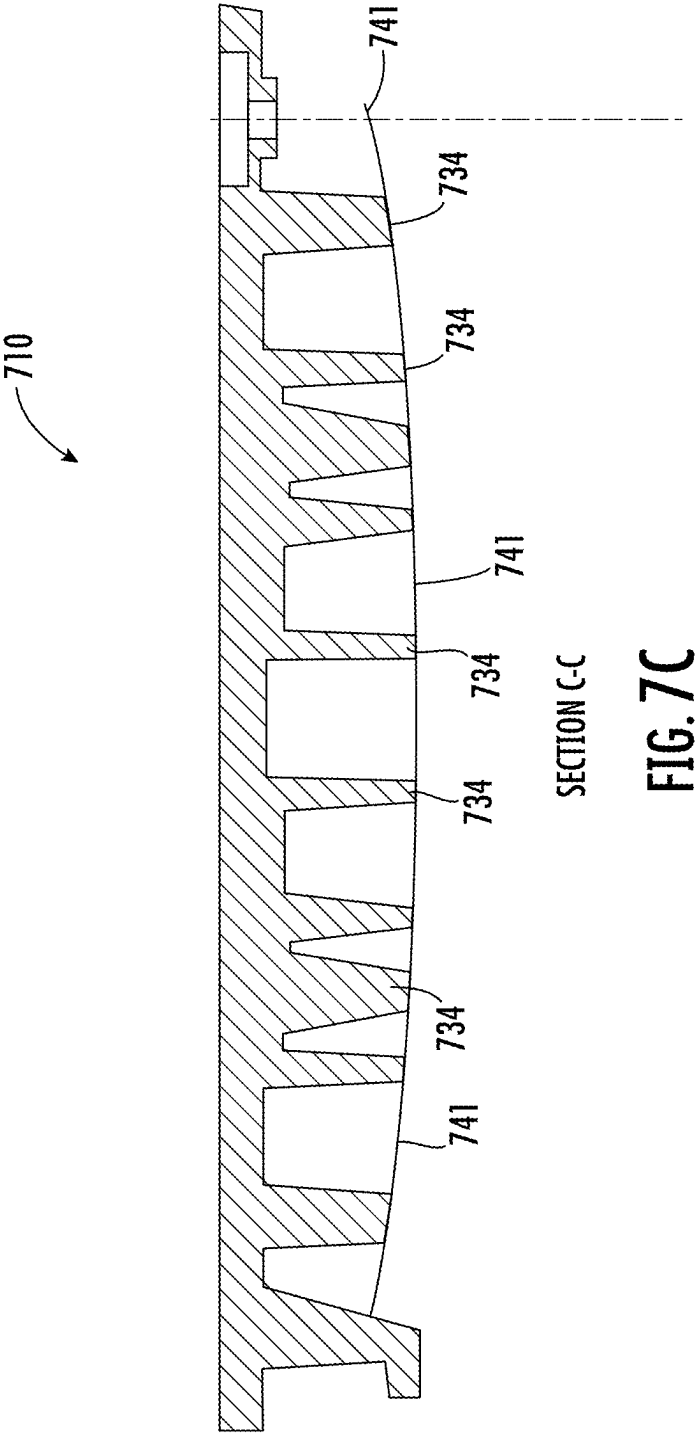
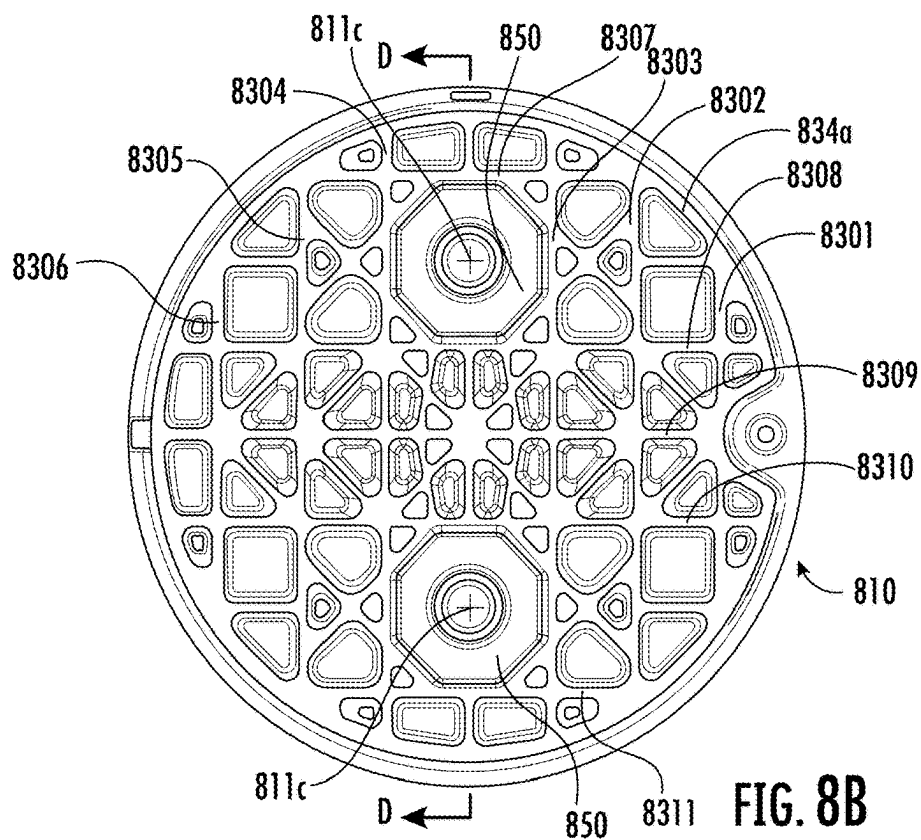
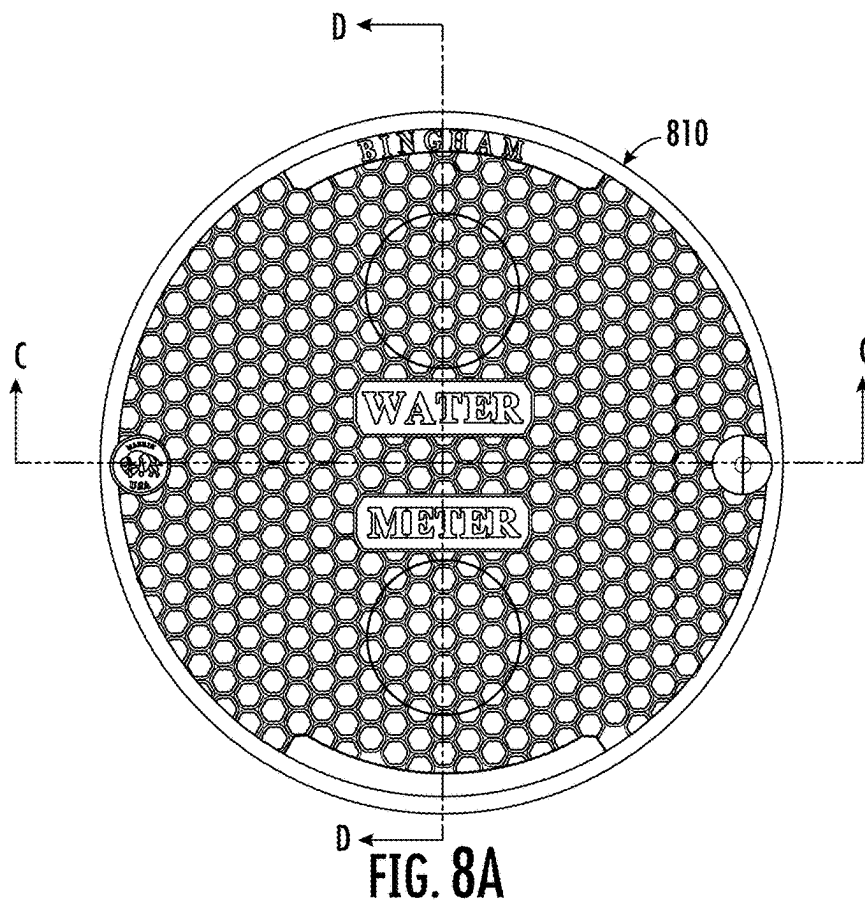
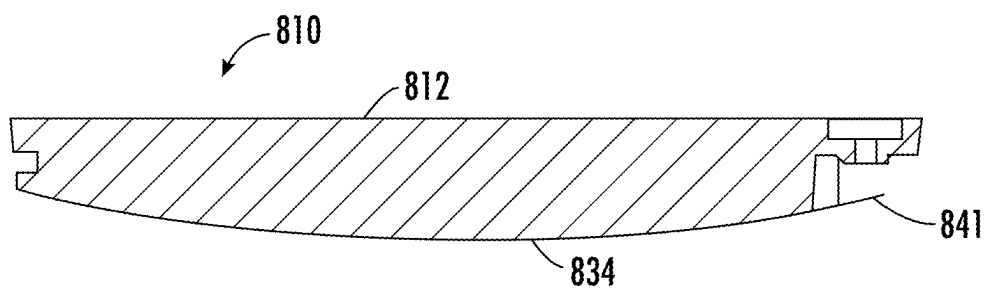


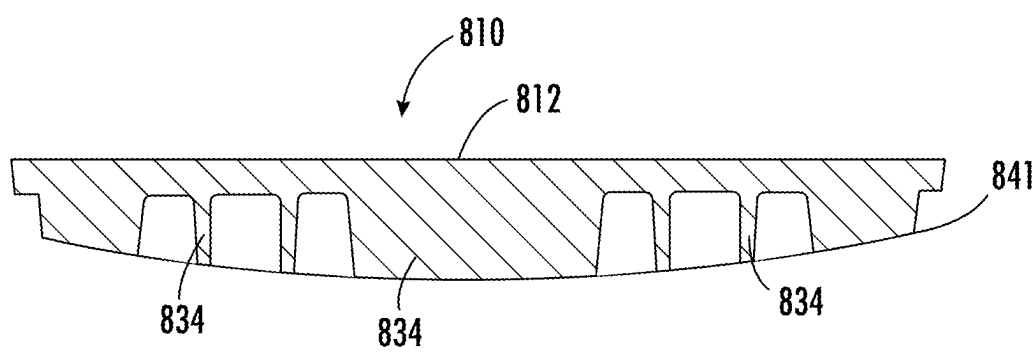
FIG. 7B







SECTION C-C
FIG. 8C



SECTION D-D
FIG. 8D

WATER OR UTILITY METER BOX LID SUITABLE FOR MULTIPLE TYPES OF LATCH MECHANISMS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims benefit of U.S. Provisional Application No. 63/556,778, filed Feb. 22, 2024, and entitled Water or Utility Meter Box Lid Suitable for Multiple Types of Latch Mechanisms the entire disclosure of which is incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

[0002] The present disclosure relates to a lid for use with a water or utility meter box, and more specifically to a water or utility meter box lid that is suitable for having installed thereon multiple different types of latch mechanisms.

BACKGROUND

[0003] Water and/or utility meter boxes are very commonly employed by, e.g., municipality entities and the like, to house instruments of varying types, water and/or gas flow meters. Such water or utility meter boxes are typically securely mounted or buried for safety, stability and/or security, but the instruments housed therein may often need to be inspected, repaired, or otherwise accessed by personnel. To help facilitate such access, such water and/or utility meter boxes are often fitted with lids that can be removed by a user.

[0004] Such lids come in a variety of different sizes and types (depending, e.g., on the type of instruments housed therein) and commonly employ latches to maintain the lid in place on the water and/or utility meter box. However, such water or utility meter box lids suffer from certain challenges, e.g., lack of interchangeability between parts, etc., that drive up costs and prevent easier installation and maintenance.

SUMMARY

[0005] The following presents a simplified summary of the claimed subject matter in order to provide a basic understanding of some aspects of the claimed subject matter. This summary is not an extensive overview of the claimed subject matter. It is intended to neither identify key or critical elements of the claimed subject matter nor delineate the scope of the claimed subject matter. Its sole purpose is to present some concepts of the claimed subject matter in a simplified form as a prelude to the more detailed description that is presented later.

[0006] The water and utility meter box lid and latch system described hereinbelow, according to various embodiments, addresses various challenges facing previously employed lid and latch systems. A particular advantage of the lid described herein is that it includes structural features that render it suitable for receiving a variety of different types of latch mechanisms, e.g., worm latches, spring-loaded latches, etc. In this way, a manufacturer of such a lid may manufacture a single type of lid to be used in the manufacturing and/or installation of a variety of different lid/latch combinations. By manufacturing a single type of lid to be used in the manufacturing and/or installation of a variety of different lid/latch combinations, the manufacturer can enjoy significant manufacturing costs savings, e.g., by requiring only a single molding tool for a given size of lid, rather than multiple molding tools, and can pass those cost

savings onto customers such as municipalities and other water and/or utility entities (and hence save taxpayer money too). In addition, by manufacturing a single type of lid to be used in the manufacturing and/or installation of a variety of different lid/latch combinations, a manufacture can provide significant cost savings for its own, as well as its customers', administrative tasks by enabling easier storage and selection of only a single lid, rather than multiple lids, and can again pass those costs savings onto customers and taxpayers. Still further, by manufacturing a single type of lid to be used in the manufacturing and/or installation of a variety of different lid/latch combinations, a customer such as a municipality or other water/utility entity can avoid confusion when selecting a suitable lid to be installed with a chosen latch mechanism, since the lid is suitable for any such chosen latch mechanism, thereby potentially avoiding or reducing installation errors and/or safety concerns. In accordance with various embodiments thereof, systems and methods are provided for a lid for a water or utility meter box that enables selection/interchangeability between different types of latch mechanisms, the lid accommodating either one of a selected worm latch, spring-loaded latch, etc.

[0007] For example, in an embodiment, there is provided a lid for covering a utility or meter box, the lid configured to accommodate either of a selected one of a worm latch or a spring-loaded latch. The lid may comprise a top side, a bottom side, and a region capable of receiving the selected one of the worm latch or the spring-loaded latch. The latch-receiving region may include a through-hole extending from the top side to the bottom side, the bottom side including a bottom surface that defines an opening configured to accommodate a spring of the spring-loaded latch.

[0008] The opening may be a spring retainer hole in the bottom surface of the lid, the spring retainer hole configured to receive an end of the spring. Additionally or alternatively, the bottom surface may have a lip coaxial with the through-hole, the lip extending downwardly from the bottom surface, the lip having a bottom lip surface spaced apart from the bottom surface, such that the opening is a cavity defined between the bottom lip surface and the bottom surface. The cavity may be configured to accommodate at least a portion of a spring.

[0009] The bottom lip surface may be configured to rotatably engage a latching portion of the selected one of the worm latch or the spring-loaded latch. In addition, the bottom side may include a plurality of ribs having a web configuration. The plurality of ribs may include a rib having a rib wall, the rib wall at least partially defining the cavity, the rib wall configured to engage at least a portion of a spring of a spring-loaded latch.

[0010] In addition, the top side may have a recess that is coaxial with the through-hole. The recess may be configured to receive an engagement portion of the worm latch or the spring-loaded latch, the engagement portion being accessible to a user from the top side of the lid to move the latch mechanism between an unlatched position and a latched position. The through-hole may define an axis generally perpendicular to the top side, the through-hole configured to house the selected one of either the worm latch or the spring-loaded latch, such that, when the through-hole houses a worm latch, the worm latch rotates around a worm-latch axis that is coaxial with the through-hole axis, and, when the through-hole houses a spring-loaded latch, the spring-loaded

latch rotates around a spring-loaded latch axis that is also coaxial with the through-hole axis.

[0011] In various embodiments, there is provided a lid for covering a utility or meter box, the lid comprising a top side, a bottom side, a through-hole extending from the top side to the bottom side, the top side having a recess at one end of the through-hole, the bottom side defining a bottom surface that has a lip that extends downwardly from the bottom surface, and the lip having a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface.

[0012] The through-hole may be configured to receive a selected one of a worm latch or a spring-loaded latch. The bottom lip surface may be configured to rotatably engage a latching portion of the selected one of the worm latch or the spring-loaded latch. The cavity may be configured to accommodate at least a portion of a spring if the selected latch is a spring-loaded latch.

[0013] The bottom side may include a plurality of ribs having a web configuration. The bottom side may include a rib having a rib wall, the rib wall at least partially defining the cavity, the rib wall configured to engage at least a portion of a spring if the selected latch is a spring-loaded latch.

[0014] In further embodiments, there is provided a lid for covering a utility or meter box, the lid comprising a top side, a bottom side; and a through-hole extending from the top side to the bottom side and defining an axis generally perpendicular to the top side, the through-hole configured to house either a worm latch or a spring-loaded latch. When the through-hole houses a worm latch, the worm latch may rotate around a worm-latch axis that is coaxial with the through-hole axis. Likewise, when the through-hole houses a spring-loaded latch, the spring-loaded latch may rotate around a spring-loaded latch axis that is also coaxial with the through-hole axis.

[0015] The top side of the lid may have a recess coaxial with the through-hole. The recess may be configured to receive an engagement portion of the worm latch or the spring-loaded latch, the engagement portion being accessible to a user from the top surface of the lid to move the latch mechanism between an unlatched position and a latched position. The bottom side may define a bottom surface and may have a lip that extends downwardly from the bottom surface. The lip may have a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface. The bottom lip surface may be configured to rotatably engage a latching portion of the worm latch or the spring-loaded latch. The cavity may be configured to accommodate at least a portion of a spring of the spring-loaded latch.

[0016] In still further embodiments, there is provided a lid for covering a utility or meter box, the lid configured to accommodate either of a selected one of a worm latch or a spring-loaded latch. The lid may comprise a top side, a bottom side that includes a plurality of ribs having a web configuration. The web configuration may include a plurality of rib rings and a plurality of chordal ribs that intersect the rib rings. The plurality of rib rings may include an outer rib ring, both ends of each one of the plurality of chordal ribs intersecting the outer rib ring. Additionally or alternatively, the plurality of rib rings may include an inner rib ring with which the plurality of chordal ribs intersect.

[0017] The plurality of chordal ribs may include at least one pair of chordal ribs that are parallel to each other. The

plurality of chordal ribs may also include at least one pair of chordal ribs that are perpendicular to each other. Additionally, the plurality of chordal ribs may include at least one pair of chordal ribs that are at a 45 degree angle relative to each other. In various embodiments, the bottom-most edge of the plurality of chordal ribs may define a generally flat plane. Alternatively, the bottom-most edge of the plurality of chordal ribs may define a curved plane.

[0018] In still additional embodiments, there is provided a method of assembling a lid for covering a utility or meter box. The method may comprise the step of storing a plurality of lids, each lid having a top side, a bottom side, and a through-hole extending from the top side to the bottom side. The method may also comprise the steps of storing at least one worm latch, the worm latch configured to be installed through the through-hole, and storing at least one spring-loaded latch, the spring-loaded latch configured to be installed through the through-hole. The method may also comprise the steps of selecting a lid from the plurality of lids and selecting a latch from the at least one worm latch or the at least one spring-loaded latch. Once selected, the method may include the step of installing the selected latch through the through-hole of the selected lid.

[0019] In such a method, the selected latch may include an engagement portion accessible to a user from the top side of the lid for moving the selected latch between an unlatched position and a latched position. The selected latch may also include a latching portion below a bottom surface on the bottom side of the lid, the latching portion configured to secure the lid to the utility or meter box when the latching portion is in the latched position.

[0020] In such method, the bottom surface may define an opening configured to accommodate a spring of the spring-loaded latch. The opening may be a spring retainer hole in the bottom surface of the lid, the spring retainer hole configured to receive an end of the spring. Additionally or alternatively, the bottom surface may have a lip coaxial with the through-hole, the lip extending downwardly from the bottom surface, the lip having a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface, the cavity for receiving at least a portion of the spring.

DRAWINGS

[0021] FIG. 1A shows a top view of a lid for a water or utility meter, in accordance with various embodiments.

[0022] FIG. 1B shows a bottom view of the lid shown in FIG. 1A, in accordance with various embodiments.

[0023] FIG. 1C shows a side cross-sectional view of the lid shown in FIG. 1A, taken along cross-section C-C of FIG. 1A, in accordance with various embodiments.

[0024] FIG. 2A is a top perspective view of a lid for a water or utility meter having a latch mechanism installed, and showing a removable cover in place within the lid, in accordance with various embodiments.

[0025] FIG. 2B is a top perspective view of the lid and latch mechanism shown in FIG. 2A, showing the removable cover removed from the lid, in accordance with various embodiments.

[0026] FIG. 2C is a bottom perspective view of the lid and latch mechanism shown in FIG. 2A, in accordance with various embodiments.

[0027] FIG. 2D is a side view of the lid and latch mechanism shown in FIG. 2A, in accordance with various embodiments.

[0028] FIG. 2E is a top view of the lid and latch mechanism shown in FIG. 2A, in accordance with various embodiments.

[0029] FIG. 2F is a side cross-sectional view of the lid shown in FIG. 2E, taken along cross-section F-F of FIG. 2E, in accordance with various embodiments.

[0030] FIG. 3A is a bottom perspective view of a lid and alternative latch mechanism, in accordance with various embodiments.

[0031] FIG. 3B is a side view of the lid and latch mechanism shown in FIG. 3A, in accordance with various embodiments.

[0032] FIG. 3C is a top view of the lid and latch mechanism shown in FIG. 3A, in accordance with various embodiments.

[0033] FIG. 3D is a side cross-sectional view of the lid shown in FIG. 3C, taken along cross-section D-D of FIG. 3C, in accordance with various embodiments.

[0034] FIG. 4A is a bottom perspective view of a lid and alternative latch mechanism, in accordance with various embodiments.

[0035] FIG. 4B is a side view of the lid and latch mechanism shown in FIG. 4A, in accordance with various embodiments.

[0036] FIG. 4C is a top view of the lid and latch mechanism shown in FIG. 4A, in accordance with various embodiments.

[0037] FIG. 4D is a side cross-sectional view of the lid shown in FIG. 4C, taken along cross-section D-D of FIG. 4C, in accordance with various embodiments.

[0038] FIG. 5A is a top view of an alternative lid, in accordance with various embodiments.

[0039] FIG. 5B is a bottom view of the alternative lid shown in FIG. 5A, in accordance with various embodiments.

[0040] FIG. 5C is a side cross-sectional view of the lid shown in FIG. 5A, taken along cross-section C-C of FIG. 5A, in accordance with various embodiments.

[0041] FIG. 6A is a top view of a still alternative lid, in accordance with various embodiments.

[0042] FIG. 6B is a bottom view of the alternative lid shown in FIG. 6A, in accordance with various embodiments.

[0043] FIG. 6C is a side cross-sectional view of the lid shown in FIG. 6A, taken along cross-section C-C of FIG. 6A, in accordance with various embodiments.

[0044] FIG. 7A is a top view of a further alternative lid, in accordance with various embodiments.

[0045] FIG. 7B is a bottom view of the alternative lid shown in FIG. 7A, in accordance with various embodiments.

[0046] FIG. 7C is a side cross-sectional view of the lid shown in FIG. 7A, taken along cross-section C-C of FIG. 7A, in accordance with various embodiments.

[0047] FIG. 8A is a top view of a still further alternative lid, in accordance with various embodiments.

[0048] FIG. 8B is a bottom view of the alternative lid shown in FIG. 8A, in accordance with various embodiments.

[0049] FIG. 8C is a side cross-sectional view of the lid shown in FIG. 8A, taken along cross-section C-C of FIG. 7A, in accordance with various embodiments.

[0050] FIG. 8D is another side cross-sectional view of the lid shown in FIG. 8A, taken along cross-section D-D of FIG. 8A, in accordance with various embodiments.

DETAILED DESCRIPTION

[0051] Reference will now be made in detail to specific embodiments illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth to provide a thorough understanding. However, it will be apparent to one of ordinary skill in the art that embodiments may be practiced without these specific details. In other instances, known methods, procedures and/or components have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

[0052] As set forth above, FIG. 1A shows a top view of a lid 10 for a water or utility meter, in accordance with various embodiments. According to an example embodiment, the lid 10 is configured for covering a utility or meter box (not shown in FIG. 1). It should be noted that FIG. 1A is merely exemplary, and thus the shape and position of the various features illustrated therein, and in the other figures, are merely examples. Additional figures, illustrating specific embodiments of various features and functionality, will be provided in further detail below.

[0053] In the embodiment shown in FIG. 1A, the lid 10 includes a top side 12. Advantageously, the top side 12 has a generally flat configuration, so as to enable it to be employed in locations over which, e.g., pedestrians, automobiles or other vehicles can move over it without interfering therewith or being interfered with thereby. Whether generally flat or having some other configuration, the top side 12 may include various shapes or features, e.g., writings, logos, strengthening ribs, or the like, that are molded therein via interchangeable tooling inserts, which may be customizable for the given use or customer. Providing interchangeable tooling inserts for the purposes of enabling such customizable features has the advantage that a common molding tool for the lid can be employed across a wide variety of customers, with only the insert being interchanged for specific customers, thereby making manufacturing of such lids easier and less expensive while still enabling customers to enjoy the benefits of having lids with their names or logos thereon. Some of these benefits may include having the utility company to which a lid belongs more easily identified, for example in situations where service areas overlap across service providers.

[0054] The lid 10 may be manufactured from a variety of different materials. For example, the lids can be manufactured from cast iron, plastic, glass reinforced polymer (e.g., using a homopolymer resin), or composite. In addition, the lids 10 may have solid and flush mount AMI/AMR antennae mount options in single or multiple positions to accommodate multiple antenna applications, and/or single or multiple threaded boss locations for underside mount antennas. Still further, the lids may include a variety of screw anchor positions for mounting various hardware, and/or press-in ports to accommodate various clips and retainers for common antennas.

[0055] FIG. 1B shows a bottom view of the lid shown in FIG. 1A, while FIG. 1C shows a side cross-sectional view of the lid shown in FIG. 1A, taken along cross-section C-C of FIG. 1A. As shown therein, the lid 10 also includes a bottom side 14. The bottom side 14 may also have a generally flat configuration, or, as shown in FIG. 1C, may be comprised of a variety of different surfaces having a variety of different shapes and situated at a variety of different levels, as will be discussed in further detail below.

[0056] Referring, for example, to FIG. 1C, the lid 10 may define a latch through-hole 18. The latch through-hole 18 extends from the top side 12 of the lid 10 to the bottom side 14 of the lid 10. The lid 10 may also include a recess 26 on the top side 12. The recess 26 may be coaxial with the through-hole 18, or may be offset relative thereto. In addition, the bottom side 14 of the lid 10 may include a bottom surface 24 in the region of the latch through-hole 18. The bottom surface 24 may have a lip 28. The lip 28 may be circular in shape and surround the through-hole 18 while extending downwardly from the bottom surface 24. The lip 28 may define a bottom lip surface 30 that is spaced apart from the bottom surface 24 of the lid 10. In this way, a cavity 32 is defined between the bottom lip surface 30 and the bottom surface 24. Still further, the bottom surface 24 of the lid 10 may also include a spring retainer hole 37, which may be employed to house one end of a spring for a spring-loaded latch mechanism, as will be described in additional detail below. It should be appreciated that, in alternative embodiments, the opening may instead be a boss or protrusion that engages the end of a spring, thereby performing the same function thereof.

[0057] Of course, it should be noted that, just as providing interchangeable tooling inserts for customizable features allows for a common molding tool for the lid to be employed across a wide variety of customers, the presently described lid may also provide these benefits for other features of the lid. For example, the lid 10 may also include interchangeable tooling inserts for different geometries of the latch through-hole 18, the cavity 32 defined between the bottom lip surface 30 and the bottom surface 24, the spring retainer hole 37, among other features. In this way, these features can be customized according to a customer's needs, while still enabling a common molding tool for the lid to be employed across a wide variety of customers, with only the insert being interchanged for specific customers. To ease manufacturing, the mold tools may be equipped with sliders that enable one or more such interchangeable tooling inserts to be placed in a different location or orientation, e.g., a first position of the slider corresponding to a first such tooling insert and a second position corresponding to a second such tooling insert.

[0058] As shown in FIGS. 1B and 1C, the bottom side 14 of the lid 10 may also have a plurality of ribs 34. The plurality of ribs 34 may provide additional loading strength to the lid 10, particular in those uses in which the lid 10 is employed in areas over which vehicles drive over it. Advantageously, the plurality of ribs 34 has a web configuration, which imparts enhanced loading strength to the lid 10.

[0059] More specifically, the plurality of ribs 34 in the embodiment shown and described in FIG. 1B has a web configuration that includes an outer rib ring 34a and an inner rib ring 34b. In addition, this web configuration of the plurality of ribs 34 also includes four pairs of chord ribs 34, each rib of the four pairs of chord ribs extending to and from the outer rib ring 34a. The first pair of chord ribs 34d are parallel to each other and are disposed on opposite sides of the instrument through-hole 11c. The second pair of chord ribs 34e are also parallel to each other and are disposed on opposite sides of the instrument through-hole 11c, with the second pair of chord ribs 34e being perpendicularly-disposed relative to the first pair of chord ribs 34d. The first pair

of chord ribs 34d and second pair of chord ribs 34e intersect each other at locations 34h where they cross the inner rib ring 34b.

[0060] In addition, the third pair of chord ribs 34f are parallel to each other and are disposed on opposite sides of the instrument through-hole 11c. Likewise, the fourth pair of chord ribs 34g are also parallel to each other and are disposed on opposite sides of the instrument through-hole 11c, with the third pair of chord ribs 34f being perpendicularly-disposed relative to the fourth pair of chord ribs 34g. The third pair of chord ribs 34f and fourth pair of chord ribs 34g also intersect each other at locations 34h where they cross the inner rib ring 34b. Thus, as shown in FIG. 1B, at each location 34h where the inner rib ring 34b is intersected by two respective chord ribs 34, such location 34h is molded so as to have ribs extending in six different directions, thereby providing a high degree of load strength.

[0061] Still further, because the third and fourth pairs of chord ribs 34f, 34g are rotated 45 degrees relative to the first and second pairs of chord ribs 34d, 34e, respective ones of the chord pairs also intersect with each other at locations 34j that are radially inwardly relative to the inner rib ring 34b. Thus, as shown in FIG. 1B, at each location 34j where two respective chord ribs 34 intersect each other, such location 34j is molded so as to have ribs extending in four different directions, thereby providing still further load strength to the lid 10.

[0062] In addition, the plurality of ribs 34 may include a specific rib wall 36 in the region of the latch through-hole 18. The rib wall 36 may at least partially define the cavity 32, providing sufficient space within the cavity to house the components of any type of latch mechanism 16, e.g., a worm latch or a spring-loaded latch, that may be employed as is described in further detail below.

[0063] As set forth above, a particular advantage of the lid 10 described herein is that it includes structural features that render it suitable for receiving a variety of different types of latch mechanisms, enabling a single type of lid to be used in the manufacturing and/or installation of a variety of different lid/latch combinations. In this regard, FIG. 2A is a top perspective view of a lid 210 for a water or utility meter having one type of latch mechanism 216 installed through the latch through-hole 218. FIG. 2A shows the lid 210 having a removable cover 211a in place within the lid 210, while FIG. 2B is a top perspective view that shows the removable cover 211a removed from its complementary recess 211b in the lid 210. As shown in FIG. 2B, the recess 211b may also define an instrument through-hole 211c through the lid 210, through which an instrument (not shown) in the water or utility meter box (also not shown) can be accessed by a user without the need to remove the entire lid 210 from the water or utility meter box.

[0064] Additional views of this lid/latch combination are shown in FIG. 2C, which illustrates a bottom perspective view of the lid and latch mechanism, FIG. 2D which illustrates a side view thereof, and FIG. 2E which illustrates a top view. Referring now to FIG. 2F, there is shown a side cross-sectional view of the lid shown in FIG. 2E, taken along cross-section F-F of FIG. 2E. In this embodiment, there is installed in the latch through-hole 218 an exemplary latch mechanism 216. In the embodiment shown, the exemplary latch mechanism 216 is a worm latch (although other types of latches could be selected and installed instead, as will be

discussed further below). This worm latch helps to illustrate the general features of any type of latch mechanism **216** that may be employed.

[0065] It should be appreciated that the latch mechanism **16** may include any type of rotating latch that, e.g., spring-centers to the latched position, any type of worm latch for worm clamp retainment, or any other mechanisms mounted on a rotating axis with, e.g., spring positioning action. Still further, it should be appreciated that the latch mechanisms employed can have different heights and/or different strokes/reaches to accommodate a wide variety of latching applications. In addition, it should be appreciated that the latch mechanism may be cast, injection molded, machined, fabricated, formed from sheet metal, stamped, or 3D printed to accommodate customization for low volume applications. Still further, the latch mechanisms may be compression-type latch mechanisms, e.g., that rotate and/or that compress from the rotating motion so as to avoid the use of an external spring, or may be a snap lock-type latch mechanism, e.g., in which the spring is contained in the body of the latch.

[0066] More specifically, as shown in FIG. 2F, the latching mechanism **216** is installed so as to extend through the latch through-hole **218**. The latching mechanism **216** includes an engagement portion **220** that is accessible to a user from the top side **212** of the lid **210**. In the embodiment shown, the engagement portion **220** is a bolt that is configured to be engaged, e.g., turned, by a complementary key so as to actuate the latching mechanism **216**. It should be recognized that the engagement portion **220** of the latch mechanism **216** may be any suitable shape or geometry which enables it to be engaged by a user, by any suitable tool for doing so. The engagement portion **220** is configured to move the latch mechanism **216** between an unlatched position and a latched position, as will be shown in further detail below.

[0067] Advantageously, and as also shown in FIG. 2F, the depth of the recess **226** is similar in height to the height of the engagement portion **220** of the latching mechanism **16** when the latching mechanism **216** is installed in the latch through-hole **218**. In this way, when the latching mechanism **216** is fully installed in the through-hole **218**, the top surface of the engagement portion **220** is flush-mounted, e.g., at a generally similar height, with the top side **212** of the lid **210**. In this way, the engagement portion **220** is less likely to interfere with vehicles or pedestrians that travel over the lid **210**.

[0068] As shown in FIG. 2F, the latching mechanism **216** also includes a latching portion **222**. The latching portion **222** extends through the latch through-hole **218** and at least some components of the latching portion **222** reside below the bottom side **214** of the lid **210**. As set forth above, the bottom side **214** of the lid **210** may include a specific bottom surface **224** in the region of the latch through-hole **218**, and the latching portion **222** may reside below such bottom surface **224** of the lid **210**. The latching portion **222** is configured to secure the lid **210** to the utility or meter box when the latching portion **222** is in the latched position, as will be shown in further detail below.

[0069] As set forth above, the bottom surface **224** has a lip **228** that extends downwardly from the bottom surface **224** to define a bottom lip surface **230**. The bottom lip surface **230** is configured to rotatably engage the latching portion **222** of the latching mechanism **216**, e.g., to have the latching portion **222** of the latching mechanism **216** rotate relative thereto, as will be shown in further detail below. The cavity

232 that is defined between the bottom lip surface **230** and the bottom surface **224** is configured to accommodate at least a portion of a spring (shown in further detail below) of a spring-loaded latching mechanism **216**, when such a spring-loaded latch mechanism is installed on the lid **210**. The rib wall **236** that at least partially defines the cavity **232** is also advantageously configured to engage at least a portion of the spring, e.g., by helping to maintain the spring in position within the cavity **232** and engaging a portion, e.g., an end, of the spring when the spring is loaded.

[0070] As set forth above, a particular advantage of the lid **10** shown herein is that it is suitable for having installed therein a variety of different types of latch mechanisms **16**. While FIGS. 2A-2F illustrate an example of the lid **210** having a worm latch installed therein, FIGS. 3A-3D and FIGS. 4A-4D illustrate examples of the lid **310** having different types of spring-loaded latches installed therein instead. It should be appreciated that the spring employed in such a spring-loaded latch mechanism could be any type of spring, e.g., torsional, extension, or compression to achieve actuation. For example, FIG. 3A is a bottom perspective view of the lid **310** having a shallow spring-loaded latch mechanism **316** installed through the latch through-hole **318**, with the latch portion **322** thereof residing in the cavity **332**. The spring-loaded type of latch mechanism **316** can also be seen in FIG. 3B, which is a side view of the lid and latch mechanism shown in FIG. 3A, and in FIG. 3D, which is a side cross-sectional view of the lid shown in FIG. 3C, taken along cross-section D-D (while FIG. 3C is a top view of the lid and latch mechanism which shows only the engagement portion **320** of the latch mechanism **316**).

[0071] As shown in FIG. 3A, the spring-loaded latch mechanism **316** also includes a spring **323**. The spring **323** is wound around the latch portion **322** and provides a spring force to maintain the latch portion **322** in the latched position unless and until a user exerts a force greater than the spring force, e.g., by engaging the engagement portion **320** of latch mechanism such as by inserting a tool and turning the engagement portion **320**, in order to move the latch portion **320** from the latched position to the unlatched position. In this embodiment, the bottom surface **324** of the lid **310** includes the spring retainer hole **337**, into which one end **323a** of the spring **323** may be inserted for maintaining the spring-loaded latch mechanism **316** in place.

[0072] FIGS. 4A-4D illustrate a further example of the lid **410** having a different type of spring-loaded latch installed therein, specifically a deep spring-loaded latch mechanism **416**. FIG. 4A is a bottom perspective view of the lid **410** having the deep spring-loaded latch mechanism **416** installed through the latch through-hole **418**, with the latch portion **422** thereof extending downwardly beyond the cavity **432**. FIG. 3B illustrates a side view of the lid **410** and deep retainer latch mechanism **416** while FIG. 3C is a top view and FIG. 3D illustrates a side cross-sectional view. Like the spring-loaded latch mechanism **316** shown in FIG. 3A, the deep retainer spring-loaded latch mechanism **416** may also include a spring **423** that is wound around the latch portion **422** and has one end **423a** thereof that is inserted into a spring retainer hole **437**. The spring **423** in this embodiment also provides a spring force to maintain the latch portion **422** in the latched position unless and until a user exerts a force greater than the spring force in order to move the latch portion **420** from the latched position to the unlatched position.

[0073] As mentioned previously, the lid 10 described herein may be manufactured in a variety of different sizes, to accommodate a wide variety of different sized water or utility meter boxes. Lids 10 of different sizes may provide different features, based on different performance characteristics needed for each. For example, larger-diameter lids may have an increased need for load strength, as compared to smaller-diameter lids, due to the distance over which it may be required to carry a load, e.g., the weight of a vehicle or the like. For this reason, the lid 10 may have different rib configurations. As set forth above, the ribs 34 on the bottom side 14 of the lid 10 advantageously have a web configuration to accommodate the load. FIGS. 5A-5D illustrate an alternative lid 510 that may be particularly suitable for a water or utility meter box, in accordance with various embodiments, that is slightly larger than the lid 10 shown and described hereinabove. For the purposes of simplicity, the discussion herein of FIGS. 5A-5D will focus on features of the lid 510 that may differ slightly from the previously-described lids (with the descriptions above of other lids providing the benefits and characteristics of the features of the lids that are similar).

[0074] Referring to FIGS. 5A-5C, FIG. 5A is a top view, FIG. 5B is a bottom view, and FIG. 5C is a side cross-sectional view of the lid shown in FIG. 5A, taken along cross-section C-C of FIG. 5A, in accordance with various embodiments.

[0075] More specifically, the plurality of ribs in the embodiment shown and described in FIG. 5B has a web configuration that includes an outer rib ring 134a and an inner rib ring 134b. In addition, this web configuration also includes four pairs of chord ribs, each rib of the four pairs of chord ribs extending to and from the outer rib ring 134a. The first pair of chord ribs 134d are parallel to each other and are disposed on opposite sides of the instrument through-hole 511c. The second pair of chord ribs 134e are also parallel to each other and are disposed on opposite sides of the instrument through-hole 511c, with the second pair of chord ribs 134e being perpendicularly-disposed relative to the first pair of chord ribs 134d. The first pair of chord ribs 134d and second pair of chord ribs 134e intersect each other at locations 134h, which are radially inwardly relative to the inner rib ring 134b.

[0076] In addition, the third pair of chord ribs 134f are parallel to each other and are disposed on opposite sides of the instrument through-hole 511c. Likewise, the fourth pair of chord ribs 134g are also parallel to each other and are disposed on opposite sides of the instrument through-hole 511c, with the third pair of chord ribs 134f being perpendicularly-disposed relative to the fourth pair of chord ribs 134g. The third pair of chord ribs 134f and fourth pair of chord ribs 134g also intersect each other at locations 134h which are radially inwardly relative to the inner rib ring 134b. Respective ones of the first and second chord ribs 134d, 134e, intersect with respective ones of the third and fourth chord ribs 134f, 134g at locations 134m along the outer rib ring 134a.

[0077] In addition to the chord ribs 134d, 134e, 134f, 134g that extend between the outer and inner rib rings 134a, 134b, the lid 510 in this embodiment also includes a plurality of connecting ribs 134k that also extend between the outer and inner rib rings 134a, 134b. In this embodiment, there are two such connecting ribs 134k spaced between each of the locations 134m, with each of the two connecting ribs 134k

extending from a location 134n on the outer rib ring 134a to the locations 134p at which the chord ribs intersect the inner rib ring 134b. Thus, as shown in FIG. 5B, each location 134p may be molded so as to have ribs extending in five different directions, thereby providing a high degree of load strength.

[0078] Still further, the lid 510 shown in FIG. 5B has additional intersection locations (other than the locations 134h at which the respective pairs of chord ribs intersect each other) that are radially inwardly relative to the inner rib ring 134b. Specifically, the lid 510 shown in FIG. 5B also has additional intersection locations 134q at which each chord rib intersects with another chord rib that extends at a 45 degree angle relative thereto. Thus, as shown in FIG. 5B, each location 134q is molded so as to have ribs extending in four different directions, thereby providing still further load strength to the lid 510. As also shown in FIG. 5B, the eight chord ribs are arranged around the instrument through-hole 511c so as to form an octagonal-shaped cavity 550. Still further, and as shown in the cross-sectional view of FIG. 5C, the bottom-most ends of the various ribs 134 are situated in a generally flat plane 141.

[0079] It should be recognized that, in various embodiments, the octagonal cavity 550 formed by the eight chord ribs around can either surround an instrument through-hole 511c, as shown in FIG. 5A, or can surround a similar region of a lid without such a through-hole. More specifically, the rib arrangement shown and described in FIG. 5A-5C, as well as in the other lid embodiments shown and described herein, can be implemented for lids either with or without through-holes, enabling the same lid molding to be used in the manufacture of such lids regardless of whether the lid is to have a through-hole or not, with only a through-hole mold insert being used to form the through-hole version. Thus, manufacturing different versions of the lids (one having a through-hole and one not having such) is made easier and less expensive by virtue of having the ribs particularly arranged around the center region in such a way that adequate space is provided for the through-hole, when such through-hole is desired to be present. Advantageously, the arrangement of the ribs around this central region of the lid, e.g., the octagonal shape of the ribs as shown in FIG. 5A-5C, are formed in this way such that the lid may exhibit comparable load-bearing strength regardless of whether the solid or through-hole version of the lid is used.

[0080] This arrangement of ribs as shown in FIGS. 5A-5C, as well as in all of the other embodiments shown herein, also provides the advantage that the rib geometry provides improved load distribution from the center of the lid to the supporting rims of the structures into which the lid is installed. In other words, traditional lid structures have the disadvantage that loads experienced at the center of the lid remain disproportionately supported by the center of the lid with inadequate support being provided at or near the outer circumference of the lid (e.g., where the lid rests on the supporting lip of the structure that the lid is covering). This occurs because the rib structures of traditional lid system are typically arranged in, e.g., grid or concentric circle-type arrangements. As a consequence, these traditional lid systems are more prone to failure when such loads are experienced. In contrast, the rib structures shown and described in the various embodiments herein, e.g., having intersecting ribs that are arranged in the manners described herein, better distribute such centrally-experienced loads to the outer regions of the lid. In this way, the lids as presently described

herein are less prone to failure and provide safer and more reliable support. As set forth above, this is an advantage that is provided by all of the different embodiments described herein.

[0081] Likewise, this arrangement of ribs as shown in FIGS. 5A-5C, as well as in all of the other embodiments shown herein, also provides the advantage that the rib geometry better divides the local stresses and distributes them within its own structure. In other words, traditional lid structures have the disadvantage that loads experienced by the lid aren't well distributed within itself and thus the local stresses caused by a load at a particular location of the lid is disproportionately experienced at that location with inadequate support being provided by the other regions of the lid. Again, this occurs because the rib structures of traditional lid system are typically arranged in, e.g., grid or concentric circle-type arrangements, that lack a sufficient number and arrangement of intersection points. As a consequence, these traditional lid systems are more prone to failure when such loads are experienced. In contrast, the rib structures shown and described in the various embodiments herein, e.g., having intersecting ribs that are arranged in the manners described herein, better distribute such local stresses loads within itself. In this way, the lids as presently described herein are less prone to failure and provide safer and more reliable support. Like above, this is an advantage is provided by all of the different embodiments described herein.

[0082] FIGS. 6A-6C illustrate an alternative lid 610 that may be particularly suitable for a still larger water or utility meter box, in accordance with various embodiments. For the purposes of simplicity, the discussion herein of FIGS. 6A-6D will focus on features of the lid 610 that may differ from the previously-described lids (with the descriptions above of other lids providing the benefits and characteristics of the features of the lids that are similar). FIG. 6A is a top view, FIG. 6B is a bottom view, and FIG. 6C is a side cross-sectional view of the lid shown in FIG. 6A, taken along cross-section C-C of FIG. 6A, in accordance with various embodiments.

[0083] More specifically, the plurality of ribs in the embodiment shown and described in FIG. 6B has a web configuration that includes an outer rib ring 634a and an inner rib ring 634b. In addition, this web configuration also includes four pairs of chord ribs, each rib of the four pairs of chord ribs extending to and from the outer rib ring 634a. The first pair of chord ribs 634d are parallel to each other and are disposed on opposite sides of the instrument through-hole 611c. The second pair of chord ribs 634e are also parallel to each other and are disposed on opposite sides of the instrument through-hole 611c, with the second pair of chord ribs 634e being perpendicularly-disposed relative to the first pair of chord ribs 634d. The first pair of chord ribs 634d and second pair of chord ribs 634e intersect each other at locations 634h, which are radially inwardly relative to the inner rib ring 634b.

[0084] In addition, the third pair of chord ribs 634f are parallel to each other and are disposed on opposite sides of the instrument through-hole 611c. Likewise, the fourth pair of chord ribs 634g are also parallel to each other and are disposed on opposite sides of the instrument through-hole 611c, with the third pair of chord ribs 634f being perpendicularly-disposed relative to the fourth pair of chord ribs 634g. The third pair of chord ribs 634f and fourth pair of chord ribs 634g also intersect each other at locations 634h

which are radially inwardly relative to the inner rib ring 634b. Respective ones of the first and second chord ribs 634d, 634e, intersect with respective ones of the third and fourth chord ribs 634f, 634g at locations 634m along the outer rib ring 634a.

[0085] In addition to the chord ribs 364d, 634e, 634f, 634g that extend between the outer and inner rib rings 634a, 634b, the lid 610 in this embodiment also includes a plurality of connecting ribs 634k that also extend between the outer and inner rib rings 634a, 634b. In this embodiment, there are two such connecting ribs 634k spaced between each of the locations 634m, with each of the two connecting ribs 634k extending from a location 634n on the outer rib ring 634a to the locations 634p at which the chord ribs intersect the inner rib ring 634b. Thus, as shown in FIG. 6B, at each location 634p is molded so as to have ribs extending in five different directions, thereby providing a high degree of load strength.

[0086] Still further, the lid 610 shown in FIG. 6B has additional intersection locations (other than the locations 634h at which the respective pairs of chord ribs intersect each other) that are radially inwardly relative to the inner rib ring 634b. Specifically, the lid 610 shown in FIG. 6B also has additional intersection locations 634q at which each chord rib intersects with another chord rib that extends at a 45 degree angle relative thereto. Thus, as shown in FIG. 6B, each location 634q is molded so as to have ribs extending in four different directions, thereby providing still further load strength to the lid 610. As also shown in FIG. 6B, the eight chord ribs are arranged around the instrument through-hole 611c so as to form a circular-shaped cavity 550 (unlike the hexagonally-shaped cavity of FIG. 5B). This circular-shaped cavity 550 provides the eight chord ribs with some additional rib material as compared to the hexagonally-shaped cavity of FIG. 5B, providing the lid with still increased load strength. As shown in the cross-sectional view of FIG. 6C, the bottom-most ends of the various ribs 634 are situated in a generally flat plane 641.

[0087] FIGS. 7A-7C illustrate a still alternative lid 710 that may be particularly suitable for a still larger water or utility meter box, in accordance with various embodiments. For the purposes of simplicity, the discussion herein of FIGS. 7A-7C will focus on features of the lid 710 that may differ from the previously-described lids (with the descriptions above of other lids providing the benefits and characteristics of the features of the lids that are similar). FIG. 7A is a top view, FIG. 7B is a bottom view, and FIG. 7C is a side cross-sectional view of the lid shown in FIG. 7A, taken along cross-section C-C of FIG. 7A, in accordance with various embodiments.

[0088] More specifically, the plurality of ribs in the embodiment shown and described in FIG. 7B has a web configuration that includes an outer rib ring 734a. In addition, this web configuration also includes six vertical (when seen in this view) chord ribs 7301, 7302, 7303, 7304, 7305, 7306, extending vertically to and from the outer rib ring 734a, each being parallel relative to each other. In addition, this web configuration also includes six horizontal (when seen in this view) chord ribs 7307, 7308, 7309, 7310, 7311, 7312, extending horizontally to and from the outer rib ring 734a, each being parallel relative to each other and each being perpendicular to the vertical-disposed chord ribs.

[0089] In addition, this web configuration also includes six downward-angled (when seen in this view) chord ribs 7313, 7314, 7315, 7316, 7317, 7318, extending from the upper left

of the figure at an angle to the lower right corner of the figure, to and from the outer rib ring **734a**, each being parallel relative to each other. In addition, this web configuration also includes six upwardly-angled (when seen in this view) chord ribs **7319**, **7320**, **7321**, **7322**, **7323**, **7324**, extending from the lower left of the figure at an angle to the upper right corner of the figure, to and from the outer rib ring **734a**, each being parallel relative to each other and each being perpendicular to the downwardly-angled chord ribs. In such an arrangement, as shown in FIG. 7B, the chord ribs intersect each other at multiple locations **734p**, multiple of such locations being molded so as to have ribs extending in six different directions, thereby providing a high degree of load strength.

[0090] Furthermore, similar to the embodiment shown in FIG. 6B, in the lid **710** of FIG. 7B, the chord ribs that are arranged directly around the instrument through-hole **711c** form a circular-shaped cavity **750**, providing these chord ribs with some additional rib material, and consequently some additional load strength, as compared to the hexagonally-shaped cavity of FIG. 5B.

[0091] Referring to the cross-sectional view of FIG. 7C, it is shown here that the bottom-most ends of the various ribs **734** are, unlike the previously-described embodiments, not situated in a generally flat plane **641**. Rather, the bottom-most ends of the various ribs **734** are situated along a curved plane **741**, such that the ribs that are located closer to the center of the lid **710** extend further than the ribs located towards the outer ring **734a**. In this way, the lid **710** may have still increased load strength.

[0092] FIGS. 8A-8C illustrate a still alternative lid **810** that, not only may be particularly suitable for a still larger water or utility meter box, but may also be suitable for a water or utility meter box that has multiple instruments therein and thus may require a lid that has multiple instrument through-holes **811c**, in accordance with various embodiments. For the purposes of simplicity, the discussion herein of FIGS. 8A-8C will focus on features of the lid **810** that may differ from the previously-described lids (with the descriptions above of other lids providing the benefits and characteristics of the features of the lids that are similar). FIG. 8A is a top view, FIG. 8B is a bottom view, FIG. 8C is a side cross-sectional view of the lid shown in FIG. 8A, taken along cross-section C-C of FIG. 8A, and FIG. 8D is a side cross-sectional view of the lid shown in FIG. 8A, taken along cross-section D-D of FIG. 8A in accordance with various embodiments.

[0093] More specifically, the plurality of ribs in the embodiment shown and described in FIG. 8B has a web configuration that includes an outer rib ring **834a**. In addition, this web configuration also includes six vertical (when seen in this view) chord ribs **8301**, **8302**, **8303**, **8304**, **8305**, **8306**, extending vertically to and from the outer rib ring **834a**, each being parallel relative to each other. In addition, this web configuration also includes five horizontal (when seen in this view) chord ribs **8307**, **8308**, **8309**, **8310**, **8311**, **8312**, extending horizontally to and from the outer rib ring **834a**, each being parallel relative to each other and each being perpendicular to the vertical-disposed chord ribs.

[0094] In addition, this web configuration also includes multiple downwardly-angled (when seen in this view) chord ribs extending from the upper left of the figure at an angle to the lower right corner of the figure, to and from the outer rib ring **834a**, each being parallel relative to each other. In

addition, this web configuration also includes multiple upwardly-angled (when seen in this view) chord ribs, extending from the lower left of the figure at an angle to the upper right corner of the figure, to and from the outer rib ring **834a**, each being parallel relative to each other and each being perpendicular to the downwardly-angled chord ribs. In such an arrangement, as shown in FIG. 8B, the chord ribs intersect each other at multiple locations **834p** having ribs extending in four, five and/or six different directions, thereby providing a high degree of load strength.

[0095] Similar to the embodiment shown in FIG. 5B, in the lid **810** of FIG. 8B, the chord ribs that are arranged directly around the instrument through-holes **811c** form hexagonal-shaped cavities **750**. In addition, referring to the cross-sectional view of FIG. 8C, the bottom-most ends of the various ribs **834** are situated along a curved plane **841**, such that the ribs that are located closer to the center of the lid **810** extend further than the ribs located towards the outer ring **834a**. In this way, the lid **810** may have still increased load strength.

[0096] In accordance with one or more of the above-described embodiments, there may also be provided a method of assembling a lid for covering a utility or meter box. As set forth above, such a method may be employed by a manufacturer of such lid/latch systems, although such a method may also be well-suited to be performed by a customer of such lid and latch products, since the system described herein provides such customer with the flexibility to use whatever latch mechanism is needed in the field regardless of the water or utility meter box that is present at the time of installation. Providing such flexibility may provide manufacturing costs savings, administrative costs savings, and/or can avoid confusion when selecting a suitable lid to be installed with a chosen latch mechanism, since the lid is suitable for any such chosen latch mechanism.

[0097] In accordance with various such embodiments thereof, the method may comprise as a first step the storing a plurality of lids, each lid having a top side, a bottom side, and a through-hole extending from the top side to the bottom side. As set forth above, the through-hole is suitable to accommodate either of a worm latch, a spring-loaded latch, etc.

[0098] The method may then comprise the steps of storing at least one worm latch, the worm latch configured to be installed through the through-hole, and storing at least one spring-loaded latch, the spring-loaded latch configured to be installed through the through-hole. Having both types of latches available enables the user the flexibility, after selecting a lid from the plurality of lids, to then select a latch from the at least one worm latch or the at least one spring-loaded latch, depending on the type of latch that the water or utility meter box requires. Once the latch is selected by the user, the method may then include the step of the user installing the selected latch through the through-hole of the selected lid.

[0099] The selected latch may include an engagement portion (described hereinabove) accessible to a user from the top side of the lid, and once the lid and latch are mated with the water or utility meter box, the user may then perform the step of moving the selected latch between an unlatched position and a latched position. The selected latch may also include a latching portion below a bottom surface on the bottom side of the lid, such that when the user performs the step of moving the selected latch between an unlatched

position and a latched position, the latching portion secures the lid to the utility or meter box.

[0100] The bottom surface of the lid may define an opening configured to accommodate a spring of the spring-loaded latch, and the user may continue the method by engaging the opening with a portion of the spring-loaded latch in order to enable the spring to perform its function of applying a spring force to the latch portion. For example, the opening may be a spring retainer hole in the bottom surface of the lid, and the user may perform the step of inserting one end of the spring into the spring retainer hole. Additionally or alternatively, the user may perform the step of positioning a portion of a spring into a cavity formed between a bottom surface of the lid and a bottom lip surface of a lip that extends downwardly from the bottom surface.

[0101] There are no limitations in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects only. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. Functionally equivalent methods and apparatuses, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. Only the terms of the appended claims are intended to be limiting, along with the full scope of equivalents to which such claims are entitled. It is also to be understood that the terminology used herein, e.g., “and”, “or”, “including”, “at least” as well as the use of plural or singular forms, etc., is for the purpose of describing examples of embodiments and is not intended to be limiting.

What is claimed is:

1. A lid for covering a utility or meter box, the lid configured to accommodate either of a selected one of a worm latch or a spring-loaded latch, the lid comprising:

a top side;

a bottom side; and

a region capable of receiving the selected one of the worm latch or the spring-loaded latch, the latch-housing region including:

a through-hole extending from the top side to the bottom side;

the bottom side including a bottom surface that defines an opening configured to accommodate a spring of the spring-loaded latch.

2. The lid of claim 1, wherein the opening is a spring retainer hole in the bottom surface of the lid, the spring retainer hole configured to receive an end of the spring.

3. The lid of claim 1, wherein

the bottom surface has a lip coaxial with the through-hole, the lip extending downwardly from the bottom surface,

the lip having a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface.

4. The lid of claim 3, wherein the cavity is configured to accommodate at least a portion of a spring.

5. The lid of claim 3, wherein the bottom lip surface is configured to rotatably engage a latching portion of the selected one of the worm latch or the spring-loaded latch.

6. The lid of claim 1, wherein the bottom side includes a plurality of ribs having a web configuration.

7. The lid of claim 6, wherein the plurality of ribs includes a rib having a rib wall, the rib wall at least partially defining the cavity, the rib wall configured to engage at least a portion of a spring of a spring-loaded latch.

8. The lid of claim 1, wherein the top side has a recess that is coaxial with the through-hole.

9. The lid of claim 8, wherein the recess is configured to receive an engagement portion of the worm latch or the spring-loaded latch, the engagement portion being accessible to a user from the top side of the lid to move the latch mechanism between an unlatched position and a latched position.

10. The lid of claim 1, wherein the through-hole defines an axis generally perpendicular to the top side, the through-hole configured to house the selected one of either the worm latch or the spring-loaded latch, such that:

when the through-hole houses a worm latch, the worm latch rotates around a worm-latch axis that is coaxial with the through-hole axis, and

when the through-hole houses a spring-loaded latch, the spring-loaded latch rotates around a spring-loaded latch axis that is also coaxial with the through-hole axis.

11. The lid of claim 1, wherein the lid includes at least one customizable feature molded therein via an interchangeable tooling insert.

12. The lid of claim 11, wherein the customizable features includes at least one of a writing, a logo, or a strengthening rib.

13. The lid of claim 1, wherein the interchangeable tooling insert is configured to change a shape, size or position of at least one of the latch through-hole, a cavity for receiving a spring latch or a spring retainer hole.

14. A lid for covering a utility or meter box, the lid comprising:

a top side;

a bottom side;

a through-hole extending from the top side to the bottom side;

the top side having a recess at one end of the through-hole;

the bottom side defining a bottom surface that has a lip that extends downwardly from the bottom surface,

the lip having a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface.

15. The lid of claim 14, wherein the through-hole is configured to receive a selected one of a worm latch or a spring-loaded latch.

16. The lid of claim 15, wherein the bottom lip surface is configured to rotatably engage a latching portion of the selected one of the worm latch or the spring-loaded latch.

17. The lid of claim 15, wherein the cavity is configured to accommodate at least a portion of a spring if the selected latch is a spring-loaded latch.

18. The lid of claim 14, wherein the bottom side includes a plurality of ribs having a web configuration.

19. The lid of claim 15, wherein the bottom side includes a rib having a rib wall, the rib wall at least partially defining the cavity, the rib wall configured to engage at least a portion of a spring if the selected latch is a spring-loaded latch.

20. A lid for covering a utility or meter box, the lid comprising:

- a top side;
- a bottom side; and
- a through-hole extending from the top side to the bottom side and defining an axis generally perpendicular to the top side, the through-hole configured to house either a worm latch or a spring-loaded latch, such that:
 - when the through-hole houses a worm latch, the worm latch rotates around a worm-latch axis that is coaxial with the through-hole axis, and
 - when the through-hole houses a spring-loaded latch, the spring-loaded latch rotates around a spring-loaded latch axis that is also coaxial with the through-hole axis.

21. The lid of claim **20**, wherein the top side has a recess coaxial with the through-hole.

22. The lid of claim **20**, wherein the recess is configured to receive an engagement portion of the worm latch or the spring-loaded latch, the engagement portion being accessible to a user from the top surface of the lid to move the latch mechanism between an unlatched position and a latched position.

23. The lid of claim **20**, wherein the bottom side defines a bottom surface and has a lip that extends downwardly from the bottom surface, the lip having a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface.

24. The lid of claim **23**, wherein the bottom lip surface is configured to rotatably engage a latching portion of the worm latch or the spring-loaded latch.

25. The lid of claim **23**, wherein the cavity is configured to accommodate at least a portion of a spring of the spring-loaded latch.

26. A lid for covering a utility or meter box, the lid configured to accommodate either of a selected one of a worm latch or a spring-loaded latch, the lid comprising:

- a top side;
- a bottom side, the bottom side including a plurality of ribs having a web configuration, the web configuration including:
 - a plurality of rib rings,
 - a plurality of chordal ribs that intersect the rib rings.

27. The lid of claim **26**, wherein the plurality of rib rings include an outer rib ring, both ends of each one of the plurality of chordal ribs intersecting the outer rib ring.

28. The lid of claim **26**, wherein the plurality of rib rings include an inner rib ring with which the plurality of chordal ribs intersect.

29. The lid of claim **26**, wherein the plurality of chordal ribs include at least one pair of chordal ribs that are parallel to each other.

30. The lid of claim **29**, wherein the plurality of chordal ribs include at least one pair of chordal ribs that are perpendicular to each other.

31. The lid of claim **29**, wherein the plurality of chordal ribs include at least one pair of chordal ribs that are at a 45 degree angle relative to each other.

32. The lid of claim **26**, wherein the bottom-most edge of the plurality of chordal ribs define a generally flat plane.

33. The lid of claim **26**, wherein the bottom-most edge of the plurality of chordal ribs define a curved plane.

34. A method of assembling a lid for covering a utility or meter box, the method comprising:

- storing a plurality of lids, each lid having a top side, a bottom side, and a through-hole extending from the top side to the bottom side;

- storing at least one worm latch, the worm latch configured to be installed through the through-hole;

- storing at least one spring-loaded latch, the spring-loaded latch configured to be installed through the through-hole;

- selecting a lid from the plurality of lids;

- selecting a latch from the at least one worm latch or the at least one spring-loaded latch;

- installing the selected latch through the through-hole of the selected lid.

35. The method of claim **34**, wherein the selected latch includes an engagement portion accessible to a user from the top side of the lid for moving the selected latch between an unlatched position and a latched position.

36. The method of claim **35**, wherein the selected latch also including a latching portion below a bottom surface on the bottom side of the lid, the latching portion configured to secure the lid to the utility or meter box when the latching portion is in the latched position.

37. The method of claim **36**, wherein the bottom surface defines an opening configured to accommodate a spring of the spring-loaded latch.

38. The method of claim **37**, wherein the opening is a spring retainer hole in the bottom surface of the lid, the spring retainer hole configured to receive an end of the spring.

39. The method of claim **37**, wherein the bottom surface has a lip coaxial with the through-hole, the lip extending downwardly from the bottom surface, the lip having a bottom lip surface spaced apart from the bottom surface so as to define a cavity between the bottom lip surface and the bottom surface, the cavity for receiving at least a portion of the spring.

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